Newcomb's Paradox as a Postselected Closed Timelike Curve

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In Newcomb's paradox an agent takes one box or two boxes. The clear box holds a small sum; a predictor has already forecast the choice and placed a large reward in the opaque box only on a forecast of one box. The paradox turns on how to formalize the correlation between choice and forecast. We physicalize that link by modeling the agent's act as a decision qubit and the predictor's accuracy as a postselected consistency factor Γ, the survival ratio of matched forecast–act pairs to mismatched ones. Under a time-travel reading the construction is an imperfect postselected consistency condition along a closed timelike curve, with the ideal P-CTC recovered at $\Gamma \to \infty$; under the laboratory reading it is ordinary postselection, with no spacetime content. When the predictor's forecast is unbiased between the choices, the verdict is classical and independent of how outcomes are scored: one-boxing is optimal once the predictor's distinguishability exceeds the reward ratio, matching the Wolpert–Benford resolution, and a classical rejection sampler reproduces every expected utility. Symmetrizing the forecast and act registers makes the setup two Newcomb problems side by side, in Lewis's sense, throughout $\Gamma \geq 1$; under per-attempt scoring the payoff ordering runs from Prisoner's Dilemma to Stag Hunt as the predictor strengthens, the crossover and risk-dominant thresholds fixed by that ratio. Beyond the payoffs, the construction leaves coherence in the act register that no classical reliability oracle with the same survival probabilities can produce, detectable by a single observable complementary to the decision basis. With the distinguishability it obeys a which-way/visibility bound, met with equality by the coherent implementation. The perfect-predictor limit dephases the register, coinciding with an act-basis measurement.

I. INTRODUCTION

William Newcomb posed the problem; Nozick [1] brought it to philosophy, and Gardner's *Scientific American* column [2] disseminated it widely. A transparent box contains a small reward m, and an opaque box contains either a large reward M or nothing. The agent chooses either to take only the opaque box, one-boxing, or to take both boxes, two-boxing. Before the choice is made, a predictor Ω has already acted: it placed M in the opaque box if it predicted one-boxing, and left the box empty if it predicted two-boxing. The predictor is assumed highly accurate. The familiar disagreement about what to do is a disagreement about how the story's correlation is represented [3–7]. Wolpert and Benford [9] locate the issue in the probabilistic structure relating choice and prediction. Lewis [8] read a Prisoner's Dilemma as a Newcomb problem, and Sobel [30] asked which Newcomb problems are Prisoner's Dilemmas. We carry that classification one type further. Starting from Newcomb's setup, the parity mechanism supplies the correlating channel Lewis stipulated, and the strength of that channel fixes the game: a Prisoner's Dilemma when the predictor is weak, a Stag Hunt when it is strong, with the crossover set by the reward ratio. Schmidt-Petri [32] reached a coordination verdict by extending to repeated play; the present route holds it at a single shot and ties the crossover to the predictor's accuracy.

We set the free-will question aside and model the act as a qubit X: its final computational-basis readout will be carried out as one-boxing or two-boxing. The "choice" is therefore a physical record produced at measurement. The predictor Ω is granted every classical fact about the setup, the preparation of X included; the single thing denied it is the realized act, which this model treats as produced at measurement. Classically, Ω can read x and implement a policy $y(x)$ without changing what x will be. With a decision qubit X, however, any mechanism that extracts the information needed to set Y is a measurement interaction in the X basis. It creates a record correlated with the eventual readout and therefore changes the state from which the later choice emerges. The predictor is not merely uncertain about a hidden classical bit; the correlating mechanism is not allowed to turn the future act into an early classical fact.

Once the predictor is forbidden from reading out X early, prediction consistency is enforced indirectly, by acting on the joint state of the relevant registers. The quantum system makes the predictor assumption operational: consistency is imposed as a joint constraint on X and Y. Closed-timelike-curve models give a language for that kind of constraint, and in a strong reading of the problem Ω may be read as a time traveler [26]. Deutsch's formulation [12] fixes a CTC marginal; for Newcomb-like situations, however, the central object is the preservation of correlations between the CTC subsystem and external registers. The postselected-teleportation approach of Lloyd *et al.* [14] conditions the joint state and so preserves correlations, avoiding the loss of correlations emphasized by Bennett *et al.* [13]. A P-CTC, in that construction, is a postselected measurement followed by renormalization. The circuit of Fig. 1 has the same structure, applied to a two-record parity check; in the time-travel reading the ancilla carries a final-state boundary condition, read as a closed timelike curve enforcing self-consistency, ideal at $\Gamma \to \infty$ and softened at finite Γ. The operational and structural claims require only a laboratory postselected parity filter and the corresponding trace-decreasing CP numerator; that this realizes a spacetime closed timelike curve is bracketed until Sec. IX. Postselection carries the computational power of PostBQP = PP [22], and whether a postselected consistency condition can be promoted from a heralded laboratory operation to a spacetime resource remains open [21].

A postselected parity check on the agent's act X and Ω's prior record Y retains agreement histories at rate p_{match} and disagreement histories at rate p_{mismatch}. Their odds ratio is the *consistency factor*,

$$\Gamma := \frac{p_{\mathrm{match}}}{p_{\mathrm{mismatch}}}, \tag{1}$$

which measures how strongly the consistency condition is enforced. At $\Gamma = 1$ the postselection supplies no consistency bias. As $\Gamma \to \infty$, mismatch histories are eliminated and the map collapses to the agreement projector. Finite $\Gamma > 1$ is the imperfect-consistency case: mismatch histories survive at relative rate Γ^{-1}. In the time-travel reading, this is the reduced two-sector parity version of an imperfect P-CTC consistency condition in the class treated by Ji et al. [19]. The payoff-ordering analysis depends on the filter only through Γ (Sec. II). When the mechanism is noisy, an unobserved spread in θ averages to an effective consistency factor Γ_{eff}, obtained from the averaged agreement and disagreement survival weights.

The construction can be illustrated by a film, much as Ji et al. [19] illustrate retrocausal communication through *Interstellar*. The pertinent film is the short *Stag Hunt* [33]. On the reading used here, a predictor relives the circumstances of a death, choosing again across a quantum multiverse where self-consistent histories survive; the two protagonists function as one looped system, each experienced as the other's counterpart, with the alive–dead swap playing the role of the in-loop operation [14] whose consistency the postselection enforces. We model that consistency as the postselected parity filter of Sec. II: the postselection retains the histories where Y agrees with X. On this reading, the room's isolation is the narrative counterpart of protecting the consistency factor from noise (Sec. VIII).

The construction supports three results. First, prediction acquires the structure of a measurement: the distinguishability $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$ and the act register's transverse visibility $\mathcal{V}$ obey $\mathcal{D}^2 + \mathcal{V}^2 \le 1$, with equality for the coherent implementation, and $\Gamma \to \infty$ operationally coincides with an act-basis measurement (Secs. II and VII). Second, at the unbiased setup prior the one-boxing verdict is classical and convention-independent: one-boxing is optimal once $\mathcal{D}$ exceeds the reward ratio m/M, matching the Wolpert–Benford resolution (Sec. III). Third, under per-attempt scoring the consistency factor then carries the induced symmetric game from Prisoner's-Dilemma to Stag-Hunt ordering (Sec. IV).

Table I maps the regimes considered and the section treating each.

Regime	prior β	scoring	key result
Payoff ordering	—	per att.	PD/SH at $\Gamma = \frac{M+m}{M}$ (Prop. D)
Unbiased verdict	$\beta = \frac{1}{2}$	both	$\Gamma = \frac{M+m}{M-m}$ (Thm. 1)
Fixed-β	exog. β	per att.	boundary Eq. (18)
Adaptive (fcst.)	$\beta_O(\alpha)$	per att.	one-box in SH (Sec. VI)
Adaptive (adv.)	$\arg\min_\beta$	per att.	$\Gamma = \frac{M+m}{m}$; Eq. (34)
Cond., fixed Y	any	on succ.	entrywise two-box dom. (App. C)
Cond., policy	β	on succ.	prior-dep.; = per att. at $\beta = \frac{1}{2}$
Noisy θ	—	per att.	$\Gamma \to \Gamma_{\mathrm{eff}}$ (Sec. VIII)

TABLE I. Map of regimes. Each row specifies a distinct decision setup through its setup-prior treatment and scoring convention. The setup-prior column is where the adaptive thresholds enter. Forward references are intentional; the table is a reading guide.

The paper is organized as follows. Section II defines the two-sector postselected consistency condition and derives the consistency factor Γ. Section III fixes the scoring conventions, proves the convention-independent verdict at the unbiased prior, and identifies it with the Wolpert–Benford resolution. Section IV computes the induced payoff ordering and the Prisoner's-Dilemma/Stag-Hunt thresholds, and Sec. V decomposes the resulting game into two Newcomb problems. Section VI states the adaptive-predictor results, with the derivations in Appendices G–I. Section VII derives the distinguishability–visibility complementarity and the σ_x signature. Section VIII describes noise through an effective consistency factor. Appendix A proves the survival-weight reduction used in Sec. II E; Appendix C gives the conditional-on-success calculation, and Appendix E gives the potential-landscape calculation used for basin selection. Proofs of the propositions stated in the body are collected in the appendices.

II. A TWO-SECTOR POSTSELECTED CONSISTENCY CONDITION

The mechanism here is a postselected parity filter: register Y represents Ω's prior record and X the agent's later act, with the postselection suppressing inconsistent histories (i.e., histories where $Y \neq X$). If ρ is the joint state and K is the Kraus operator for the successful postselection outcome, the update is

$$\rho \longmapsto \frac{K\rho K^{\dagger}}{\mathrm{Tr}(K\rho K^{\dagger})}. \tag{2}$$

Equation (2) is the postselected update of the P-CTC construction (Sec. I). The payoff-ordering results below use only the trace-decreasing map and its sector survival weights; the P-CTC reading is one available interpretation. The predictor's epistemic situation, stated informally in the introduction, is used throughout, so we record it once:

Assumption 1 (the predictor's epistemic access)**.** Ω *knows every classical fact about the setup, including the preparation of the act register. It does not possess the realized act, which is produced only at the final measurement.*

Later sections invoke Assumption 1 wherever Ω conditions on the agent's preparation (Secs. VI and VII).

A. Circuit

An ancilla qubit a is prepared in $|0\rangle$. The circuit writes the parity $x \oplus y$ into a, applies a rotation $R_y(\theta)$, and postselects on measuring a as $|0\rangle$. The rotation is

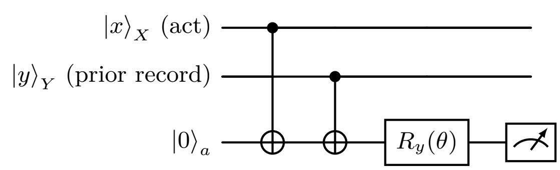


FIG. 1. The diagram represents the effective postselected map. The ancilla records the parity $x \oplus y$, is rotated by $R_y(\theta)$, and the branch is retained only when the ancilla outcome is $|0\rangle$. Agreement and disagreement histories survive with survival weights $\cos^2(\theta/2)$ and $\sin^2(\theta/2)$, respectively.

$$R_y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \tag{3}$$

equivalently $R_y(\theta) = e^{-i\theta\sigma_y/2}$. After the two CNOTs, the ancilla carries the parity: it is $|0\rangle$ on agreement and $|1\rangle$ on disagreement. Postselecting immediately on $a = |0\rangle$ would eliminate disagreement entirely. The rotation $R_y(\theta)$ softens this into a tunable consistency check: it sets the overlap between the postselected outcome and each parity sector, allowing disagreement to survive at relative rate Γ^{-1}. In the postselected-teleportation construction [14], the consistency condition is conditioning on a selected Bell outcome; here the Bell-sector condition is replaced by the parity decomposition $\mathcal{H}_{XY} = \mathcal{H}_= \oplus \mathcal{H}_{\neq}$. The ideal limit $\Gamma \to \infty$ collapses the Kraus map to the agreement projector $\Pi_=$. That construction applies the Bell projection to an in-loop NOT, where no consistent history exists and the amplitude vanishes; the parity filter applies $\Pi_=$ to a loop whose consistent histories survive and keep the map normalizable. The two constructions therefore place tunability at different points: the former obtains a normalizable map by softening the in-loop unitary toward the strict NOT, while the present circuit keeps the unitary sharp and softens the postselection through $R_y(\theta)$ on the ancilla. Incoherent noise during the would-be CTC channel can be represented as run-to-run randomness in θ (Sec. VIII).

B. Acceptance probabilities

Let A (for "acceptance") denote the event that the ancilla measurement returns 0, so that the branch is retained. The acceptance amplitude is $\langle 0|R_y(\theta)|x \oplus y\rangle$. Hence

$$\begin{aligned} p_{\text{match}}(\theta) &:= \mathbb{P}(\mathsf{A} \mid x = y) = \left|\langle 0|R_y(\theta)|0\rangle\right|^2 \\ &= \cos^2(\theta/2), \end{aligned} \tag{4}$$

$$\begin{aligned} p_{\text{mismatch}}(\theta) &:= \mathbb{P}(\mathsf{A} \mid x \neq y) = \left|\langle 0|R_y(\theta)|1\rangle\right|^2 \\ &= \sin^2(\theta/2). \end{aligned} \tag{5}$$

We refer to the pair $(p_{\text{match}}, p_{\text{mismatch}})$ as the *sector survival weights* of the filter, the survival probabilities of the agreement sector $\mathcal{H}_=$ and disagreement sector $\mathcal{H}_{\neq}$ under postselection. The match-to-mismatch ratio ratio,

$$\Gamma \;:=\; \frac{p_{\text{match}}}{p_{\text{mismatch}}} = \cot^2(\theta/2), \tag{6}$$

summarizes the strength of the filter. Γ is the consistency factor introduced in Eq. (1). The ordering results below depend on the mechanism only through Γ; a different physical implementation with the same Γ induces the same normal form.

C. Kraus operator for the postselected map

Conditioning on the ancilla outcome $|0\rangle$ yields a trace-decreasing completely positive map on XY with Kraus operator $K(\theta) = \langle 0|_a \, R_y(\theta) \, \mathrm{CNOT}_{Y\to a} \, \mathrm{CNOT}_{X\to a} \, |0\rangle_a$. Evaluating on computational basis states gives

$$\begin{aligned} K(\theta) = \; & \cos(\theta/2)\big(\,|00\rangle\langle 00| + |11\rangle\langle 11|\,\big) \\ & - \sin(\theta/2)\big(\,|01\rangle\langle 01| + |10\rangle\langle 10|\,\big). \end{aligned} \tag{7}$$

For any input ρ_{XY}, the subnormalized postselected state is $\tilde{\rho}_{XY} = K \rho_{XY} K^\dagger$ with success probability $p = \mathrm{Tr}(\tilde{\rho}_{XY})$, and conditional state $\rho'_{XY} = \tilde{\rho}_{XY}/p$. This realizes the postselected update of Eq. (2). On computational-basis inputs the map reweights the match and mismatch sectors before renormalization; on coherent inputs it also applies the relative sign visible in Eq. (7). Thus $K(\theta)$ defines a trace-decreasing completely positive numerator $\mathcal{K}_\theta(\rho) := K(\theta)\,\rho\,K^\dagger(\theta)$ on XY, with physical state update $\rho \mapsto \mathcal{K}_\theta(\rho)/\,\mathrm{Tr}[\mathcal{K}_\theta(\rho)]$. This is a two-record instance of the CP-numerator-plus-renormalization structure studied in the retrocausal-channel literature [18, 19]. In the time-travel reading, $\mathcal{K}_\theta$ is a postselected mechanism with a final-state boundary condition on the ancilla, a reduced two-sector instance of the postselected-teleportation maps treated by Ji et al. [19]. Section VIII relates this CP numerator to their noisy-P-CTC formalism; the scalar Γ remains the parity-sector survival ratio controlling the decision problem. The sector decomposition of the effect makes the correspondence concrete. Let

$$\Pi_= := |00\rangle\langle 00| + |11\rangle\langle 11|\,, \qquad \Pi_{\neq} := |01\rangle\langle 01| + |10\rangle\langle 10|\,. \tag{8}$$

The postselection effect is

$$E(\theta) := K^\dagger(\theta)K(\theta) = p_{\mathrm{match}}\,\Pi_= + p_{\mathrm{mismatch}}\,\Pi_{\neq}. \quad (9)$$

Since E acts as $p_{\mathrm{match}}\,I$ on $\mathcal{H}_=$ and $p_{\mathrm{mismatch}}\,I$ on $\mathcal{H}_{\neq}$, the consistency factor is $\Gamma = \|E|_{\mathcal{H}_=}\|/\|E|_{\mathcal{H}_{\neq}}\| = p_{\mathrm{match}}/p_{\mathrm{mismatch}}$, the likelihood ratio between the two sectors. Successful postselection multiplies the prior agreement odds by this ratio (Eq. (11)). The basis-diagonal payoff analysis depends on E and on Γ; the phase convention in K does not enter for computational-basis readouts. Off-diagonal observables such as the σ_x signature of Sec. VII A do depend on this phase.

D. Reliability in the accepted ensemble

Suppose X and Y are chosen independently with $\Pr(X = 0) = \alpha$ and $\Pr(Y = 0) = \beta$. Before postselection,

$$\Pr(x = y) = \alpha\beta + (1-\alpha)(1-\beta),$$
$$\Pr(x \neq y) = \alpha(1-\beta) + (1-\alpha)\beta.$$

Conditioned on success, the probability of agreement is

$$\Pr(x{=}y \mid \mathsf{A}) = \frac{p_{\mathrm{match}}\ \Pr(x{=}y)}{p_{\mathrm{match}}\ \Pr(x{=}y) + p_{\mathrm{mismatch}}\ \Pr(x{\neq}y)}. \quad (10)$$

Equivalently, successful postselection updates the odds of agreement by

$$\frac{\Pr(x = y \mid \mathsf{A})}{\Pr(x \neq y \mid \mathsf{A})} = \Gamma\,\frac{\Pr(x = y)}{\Pr(x \neq y)}. \quad (11)$$

Equation (11) establishes the statistical meaning of Γ: it is the multiplicative evidence supplied by the consistency condition in favor of agreement. Small θ makes $p_{\mathrm{mismatch}} \ll p_{\mathrm{match}}$, and the accepted runs are overwhelmingly consistent. At uniform input ($\alpha = \beta = 1/2$) Eq. (10) reduces to $\Pr(Y{=}X \mid \mathsf{A}) = \Gamma/(\Gamma + 1)$, so $\Gamma = 1$ gives chance and $\Gamma \to \infty$ gives certainty. Equivalently, the accepted ensemble assigns the record a *distinguishability*, the advantage over chance for inferring the act X from the record Y,

$$\mathcal{D} \;:=\; 2\Pr(Y{=}X \mid \mathsf{A}) - 1 \;=\; \frac{\Gamma - 1}{\Gamma + 1} \quad (12)$$

at the unbiased preparation, a one-to-one function of Γ mapping $[1, \infty)$ onto $[0, 1)$: at $\mathcal{D} = 0$ the record is no better than a coin, at $\mathcal{D} \to 1$ it fixes the act. Section III states the decision verdict in terms of $\mathcal{D}$, and Sec. VII identifies it with the which-path distinguishability of the act register.

E. Basis-diagonal payoff comparisons depend only on Γ

Proposition. Let $\mathcal{M}(\rho) = K\rho K^\dagger$ be a trace-decreasing CP map on $\mathcal{H}_X \otimes \mathcal{H}_Y = \mathbb{C}^2 \otimes \mathbb{C}^2$ implementing a parity check on (X, Y): (i) K is diagonal in the computational basis ($K\,|xy\rangle = K_{xy}\,|xy\rangle$) and (ii) symmetric under joint bit-flip ($(\sigma_x \otimes \sigma_x) \cdot K \cdot (\sigma_x \otimes \sigma_x) = K$, with σ_x the Pauli bit-flip).
Then $K = k_=\Pi_= + k_{\neq}\Pi_{\neq}$ with $k_= := K_{00} = K_{11}$ and $k_{\neq} := K_{01} = K_{10}$. The effect $E = K^\dagger K$ equals $p_{\mathrm{match}}\Pi_= + p_{\mathrm{mismatch}}\Pi_{\neq}$ with $p_{\mathrm{match}} = |k_=|^2$ and $p_{\mathrm{mismatch}} = |k_{\neq}|^2$. For any product input $\rho_X \otimes \rho_Y$ diagonal in the $\{C, D\}$ basis and any payoff observable diagonal in the same basis, the binary policy comparison between any two preparations α, α' depends on K only through $\Gamma := p_{\mathrm{match}}/p_{\mathrm{mismatch}}$.

Proof. See Appendix A. □

This proposition is the basis for the convention-independent verdict (Sec. III) and scopes the σ_x signature (Sec. VII A).

The basis-diagonal hypothesis is necessary: the σ_x observable measured on the symmetric preparation in Sec. VII A is not basis-diagonal, so this proposition does not constrain it; its Γ-dependence separates the coherent construction from a classical rejection sampler.

The decision analysis may then be read where agreement passes losslessly: $K = \Pi_= + \Gamma^{-1/2}\Pi_{\neq}$ with $p_{\mathrm{match}} = 1$, $p_{\mathrm{mismatch}} = 1/\Gamma$, only disagreement histories discarded. The symmetric realization $R_y(\theta)$ carries the same Γ and supplies the coherence signature of Sec. VII A and the noise model of Sec. VIII.

III. SCORING CONVENTIONS AND THE INVARIANT VERDICT

The consistency condition acts on histories before any payoff is read, and its strength is set by Γ alone (Sec. II). How the consistency-weighted histories enter the agent's expected utility is a separate question, fixed by a scoring convention.

Two conventions are available. *Per-attempt* scoring is the expected utility of a single application of the consistency condition: one run consists of preparing the act register under the agent's policy, setting the prior-record register, applying the parity filter, and reading the payoff against the consistency-weighted state $K\rho K^\dagger$. A retained run pays U and a discarded run is valued at zero, so the quantity is the single-shot expectation $\mathrm{Tr}[U\,K\rho K^\dagger] = \Pr(\mathsf{A})\,\mathbb{E}[U \mid \mathsf{A}]$; here "attempt" denotes one application of the condition. *Conditional* scoring is $\mathbb{E}[U \mid \mathsf{A}]$, the expected utility restricted to accepted runs. The two sectors carry Newcomb's two kinds of outcome. Match ($x = y$) is where Ω's record agrees with the act. Mismatch ($x \neq y$) is the case Newcomb's perfect-predictor stipulation eliminates: Ω's record disagrees with the act, so the two-boxer finds the opaque box filled and the one-boxer finds it empty. The postselection plays the role of Newcomb's predictor: rejected branches are the scenarios the predictor eliminates, kept only at relative rate Γ^{-1} for mismatch. The accepted set then contains every match scenario together with a Γ^{-1} fraction of mismatch scenarios, the latter vanishing in the perfect-predictor limit $\Gamma \to \infty$. If a joint profile (x, y) carries acceptance weight p_{xy} (so $p_{xy} = p_{\text{match}}$ when $x = y$ and $p_{xy} = p_{\text{mismatch}}$ when $x \neq y$), the per-attempt payoff is

$$u_i^{\text{eff}}(x, y) = p_{xy}\, u_i(x, y), \tag{13}$$

the unnormalized (consistency-weighted) expectation. It keeps the suppression of inconsistent histories attached to the payoff, which is what makes the temptation profile rare per attempt.

The two conventions answer different questions about the same map. Per-attempt scoring is linear in the agent's preparation, retains the discard rate, and is therefore the measure that an experimental test of the consistency condition records [14]; it is the natural measure in a no-collapse or laboratory-postselection reading, where the rejected branches occur but go unselected. Conditional scoring renormalizes, becomes nonlinear in the preparation, and is the measure available to an agent who already finds themselves in an accepted branch; the normalized ideal-P-CTC reading naturally uses it, since there the inconsistent histories have zero support. The discarded-run value zero is a definitional feature of per-attempt scoring. The two conventions agree at $\beta = 1/2$ and separate once an asymmetric base rate lets the preparation move the discard rate (Theorem 1); the verdict of this section is stated where they agree, so nothing in it depends on adjudicating between them. The game-theoretic structure of Secs. IV and V is a theorem about per-attempt scoring, and Appendix C gives the conditional calculation.

A. Base payoffs

Let m denote the transparent-box reward and M the potential opaque-box reward, with $M \gg m$ in the standard Newcomb setup [2]. Write $x \in \{C, D\}$ for the agent's action (one-box C vs two-box D), and $y \in \{C, D\}$ for the prior-record variable (the predictor's setup). The choosing role receives the standard Newcomb payoffs

$$\begin{aligned} u_1(C, C) &= M, \qquad & u_1(D, C) &= M + m, \\ u_1(C, D) &= 0, \qquad & u_1(D, D) &= m. \end{aligned} \tag{14}$$

We label Y so that $y = C$ means *predict* $X = C$ (opaque box filled) and $y = D$ means *predict* $X = D$ (opaque box empty).

B. Ensemble payoff at a fixed prior

The per-attempt utility is an ensemble statistic: the average payoff over the statistical ensemble of histories generated by the postselected joint distribution. Writing the ensemble as N samples, the accumulated payoff is

$$W_N = \sum_{k=1}^{N} \mathsf{A}_k\, u(x, y_k), \tag{15}$$

where $\mathsf{A}_k \in \{0, 1\}$ is the postselection outcome on sample k. The ratio W_N/N is the ensemble average, the same single-shot expectation $\mathrm{Tr}[U\, K\rho K^\dagger]$ defined above, now read as an ensemble mean. The ensemble is a statistical device for evaluating that expectation; no sequence of physical trials is consumed.

Let

$$\beta := \Pr(Y = C)$$

be the preselection probability that the predictor's record is C before the consistency filter acts; β is the predictor's unconditional base rate for recording C. The accuracy that defines Newcomb's predictor is carried by Γ, and β sets only the record's base rate: $\beta = 1/2$ encodes an unbiased record, and a strongly accurate predictor is a large Γ at any β. This subsection treats x as a fixed classical policy across runs; the adaptive case, in which the preparation is a decision qubit and β responds to it, is Sec. VI.

For a one-boxer ($x = C$ on every run), only the $y = C$ branch contributes:

$$\bar{u}(C) = \beta\, p_{\text{match}}\, M. \tag{16}$$

For a two-boxer ($x = D$),

$$\bar{u}(D) = \beta\, p_{\text{mismatch}}(M + m) + (1 - \beta)\, p_{\text{match}}\, m. \tag{17}$$

One-boxing has higher ensemble average iff $\bar{u}(C) > \bar{u}(D)$.

Proposition (Prior-dependent payoff boundary)**.** *For preselection setup prior $\beta \in (0, 1]$ and $M > m > 0$, one-boxing has the larger ensemble-average payoff per attempt iff*

$$\Gamma \;>\; \frac{\beta(M + m)}{\beta M - (1 - \beta) m}, \tag{18}$$

provided $\beta M > (1 - \beta) m$; otherwise one-boxing is never payoff-optimal, even in the perfect-consistency limit.

The condition $\beta M > (1-\beta) m$ is the viability condition for one-boxing in this ensemble comparison: if it fails, even the perfect-consistency limit $\Gamma \to \infty$ does not make one-boxing have the larger ensemble-average payoff. At the unbiased setup prior $\beta = 1/2$ the boundary (18) reduces to $\Gamma > (M + m)/(M - m)$. The theorem below shows that this verdict is convention-independent, and Sec. IV B identifies the same threshold with risk dominance of the one-boxing equilibrium.

C. The invariant verdict at the unbiased prior

At $\beta = 1/2$ the agent's preparation has no effect on the acceptance rate, and the two scoring conventions return the same verdict.

Theorem 1 (acceptance invariance and a convention-independent verdict)**.** *Fix $M > m > 0$ and $\Gamma \neq 1$.*

Prepare the act register with one-boxing Born weight $\alpha = \Pr(X = C)$ *and the prior record with* $\Pr(Y = C) = \beta$*, and let the parity filter act with sector survival weights* $(p_{\rm match}, p_{\rm mismatch})$*,* $\Gamma = p_{\rm match}/p_{\rm mismatch}$*.*

(i) *(Acceptance invariance.) The per-attempt acceptance probability is*

$$\Pr(\mathsf{A} \mid \alpha, \beta) = p_{\rm match}\,[\alpha\beta + (1-\alpha)(1-\beta)] + p_{\rm mismatch}\,[\alpha(1-\beta) + (1-\alpha)\beta],$$

so that

$$\frac{\partial \Pr(\mathsf{A})}{\partial \alpha} = (p_{\rm match} - p_{\rm mismatch})(2\beta - 1).$$

At $\beta = 1/2$*,* $\Pr(\mathsf{A}) = (p_{\rm match}+p_{\rm mismatch})/2$ *for every* α*, so the agent's preparation exerts no influence on the acceptance rate.*

(ii) *(Convention independence.) At* $\beta = 1/2$ *the per-attempt comparison* $\bar{u}(C)$ *versus* $\bar{u}(D)$ *and the conditional-on-success comparison* $U_{\mathsf{A}}(C)$ *versus* $U_{\mathsf{A}}(D)$ *(the expected utility given acceptance, App. C) have the same sign, both favoring one-boxing iff*

$$\Gamma > \frac{M+m}{M-m}.$$

(iii) *(Decision-theoretic match.) With accepted-ensemble reliability* $q = \Pr(Y{=}X \mid \mathsf{A}) = \Gamma/(\Gamma+1)$*, the boundary in (ii) is* $q > (M+m)/2M$*, matching the Wolpert–Benford resolution at symmetric forecast accuracy.*

Proof. Part (i) is the displayed differentiation; parts (ii) and (iii) compare the per-attempt and conditional payoffs at $\beta = 1/2$ and substitute $q = \Gamma/(\Gamma+1)$, giving $\Gamma > (M+m)/(M-m)$. See App. B. □

Remark (why $\beta = 1/2$ is the unbiased setup). At $\beta = 1/2$ the acceptance rate $(p_{\rm match} + p_{\rm mismatch})/2$ is independent of the agent's preparation, so the two scoring conventions agree (part i). The one-boxing advantage above $(M+m)/(M-m)$ then comes entirely from the record–act correlation aligning the agent's payoff with the surviving sector. For $\beta \neq 1/2$ the preparation moves the acceptance rate, the conventions separate, and the Simpson reversal of Sec. IX B appears. We take $\beta = 1/2$ as the symmetric reference case because the preparation does not move the acceptance rate there, the condition under which Nozick's agent cannot change the contents of an already-filled box; an exogenous β would introduce a base-rate bias the original setup does not stipulate.

The boundary of Theorem 1 has a compact statement in the distinguishability coordinate of Eq. (12). Converting $\Gamma = (M+m)/(M-m)$ through $\mathcal{D} = (\Gamma-1)/(\Gamma+1)$ gives the convention-independent verdict at $\beta = \frac{1}{2}$:

$$\textbf{one-box} \iff \mathcal{D} > \frac{m}{M}, \tag{19}$$

the agent one-boxes when the predictor's distinguishability exceeds the reward ratio. At the classic million-to-thousand stakes ($m/M = 10^{-3}$) this threshold is $\mathcal{D} = 0.1\%$, so a predictor only 1.048596% distinguishable from a coin flip clears it tenfold. The payoff-ordering threshold of Sec. IV sits just below it at $\mathcal{D} = m/(2M+m)$, and the adversarial adaptive threshold of Sec. VI at $\mathcal{D} = M/(M+2m)$; Fig. 4 places the three on the $\mathcal{D}$ axis.

The verdict is classical. By the proposition of Sec. II E, every quantity entering the comparison depends on the mechanism only through the sector survival weights, and a rejection sampler that keeps agreement profiles with probability $p_{\rm match}$ and disagreement profiles with probability $p_{\rm mismatch}$ reproduces all of them. The quantum content lives off the payoff basis, a payoff-independent probe of whether the consistency condition was realized coherently; Sec. VII locates it.

D. Relation to the Wolpert–Benford resolution

Wolpert and Benford resolve Newcomb's paradox by showing that the two conflicting recommendations correspond to two probabilistic structures relating the player's choice x and the predictor's forecast y [9]. Their Fearful structure decomposes $P(x,y) = P(y \mid x)\,P(x)$, treating forecast accuracy $P(y \mid x)$ as given; their Realist structure decomposes $P(x,y) = P(x \mid y)\,P(y)$ with $P(y)$ fixed by the predictor and the agent's strategy a y-independent conditional $P(x \mid y) = h(x)$, so the choice does not depend on the forecast. The two cannot be merged: specifying both conditional families simultaneously overconstrains the joint (§3.5 of [9]). WB rest their resolution on this incompatibility and set the accuracy parameter aside; we reinstate it here to surface the threshold it controls within their Fearful structure.

The accepted-ensemble reliability $q := \Pr(Y{=}X \mid \mathsf{A}) = \Gamma/(\Gamma+1)$ (Sec. II D) maps onto Fearful's symmetric accuracy parameter. At $\Gamma = 1$, $q = 1/2$: the postselection induces no record–act dependence, and the accepted joint is the product distribution, the Realist assignment. As $\Gamma \to \infty$, $q \to 1$: the accepted joint approaches Fearful's perfect-predictor limit $P(y \mid x) = \delta_{y,x}$.

Reinstating a symmetric accuracy q in the Fearful structure, the expected-utility comparison is

$$\begin{aligned} \mathrm{EU}(\text{one-box}) &= Mq, \\ \mathrm{EU}(\text{two-box}) &= (M+m)(1-q) + mq = M + m - Mq, \end{aligned} \tag{20}$$

so one-boxing wins iff $q > (M+m)/(2M)$. Within the Fearful structure WB note that the optimal choice is determined once the accuracy is fixed (their §3.1), but treat its value as immaterial to dissolving the paradox; the boundary made explicit here is that accuracy threshold. Converting the threshold of Theorem 1 through $q =$

$\Gamma/(\Gamma+1)$:

$$q \;>\; \frac{(M+m)/(M-m)}{(M+m)/(M-m)+1} \;=\; \frac{M+m}{2M},$$

the same condition. The Fearful accuracy requirement made explicit here therefore coincides with the convention-independent verdict at $\beta = 1/2$, and Sec. IV B identifies both with risk dominance of the one-boxing equilibrium.

The lower payoff-ordering threshold $(M+m)/M$, where the Stag-Hunt equilibrium first appears (Sec. IV), is additional game structure that their pure decision-theory treatment does not resolve; the consistency factor supplies this finer threshold as well.

The construction does not merge the Fearful and Realist structures: it parameterizes the Fearful family through the postselection collider and meets the Realist joint only at $\Gamma = 1$. No single value of Γ instantiates both decompositions simultaneously, so the no-merge theorem is not violated. That the decision content is entirely classical is consistent with their conclusion that quantum mechanics is not essential to the paradox. The two decompositions are the two time directions of the conditional. Realist conditions the act on a fixed prior forecast, the forward direction: the act cannot depend on it, and the joint is the product at $\Gamma = 1$. Fearful conditions the forecast on the act, the backward direction, with the record tracking the later choice once $\Gamma > 1$. Their time-reversal-invariance result (§3.4 of [9]), that the paradox is unchanged if prediction occurs after the choice, supports the reading-relative treatment of the collider/common-cause contrast in Sec. IX B (Fig. 2).

IV. THE INDUCED GAME: FROM PRISONER'S DILEMMA TO STAG HUNT

Under per-attempt scoring the verdict of Theorem 1 acquires an anatomy: the consistency weights fold into the payoff table, and the resulting normal form changes ordinal type as Γ grows. This section and the two that follow are downstream of the measurement structure: the game theory is the shape the distinguishability $\mathcal{D}$ takes under one scoring convention, the thresholds of Sec. III reappearing as equilibrium boundaries. The agreement filter is symmetric between the two registers, so the induced normal form is symmetric, and we use the standard 2×2 notation

$$\begin{array}{c|cc} & C & D \\ \hline C & (R,R) & (S,T) \\ D & (T,S) & (P,P) \end{array} \qquad (21)$$

to classify the effective payoff ordering. The symmetric table is a bookkeeping device for the two-register payoff ordering: the register Y is still Ω's prior record, and equilibrium-selection language becomes literal only once Y is promoted to a second decision register (Sec. V).

The symmetrized payoff table has Prisoner's-Dilemma ordering when

$$T > R > P > S,$$

so D strictly dominates C and (D, D) is the unique diagonal fixed point (a Nash equilibrium of the symmetric game, Sec. V). It has the Stag-Hunt structure [11] when

$$R > T, \qquad R > P, \qquad P > S,$$

so both (C, C) and (D, D) are diagonal fixed points.

If one further distinguishes strict ordinal payoff chains, then the coordination region splits into two subregimes:

$$R > T > P > S$$

is the strict ordinal Stag-Hunt chain, while

$$R > P > T > S$$

is a strong-assurance, or diagonal-dominant, coordination regime. This second regime has the same two diagonal fixed points, but both matched outcomes dominate both mismatched outcomes.

At $\Gamma = 1$, the postselection gives no consistency bias and the two registers are weighted symmetrically. As Γ increases, the survival factors raise R relative to T, and the ordering passes from $T > R$ (Prisoner's Dilemma) to $R > T$ (Stag Hunt) at a threshold value of Γ. These inequalities classify the matrix only; equilibrium selection requires the Newcomb payoffs and the setup prior (Sec. IV B).

For each computational-basis profile (x, y), the postselection effect supplies the survival weight

$$p_{xy} = \langle xy|\, E(\theta)\, |xy\rangle = \begin{cases} p_{\text{match}}, & x = y, \\ p_{\text{mismatch}}, & x \neq y. \end{cases}$$

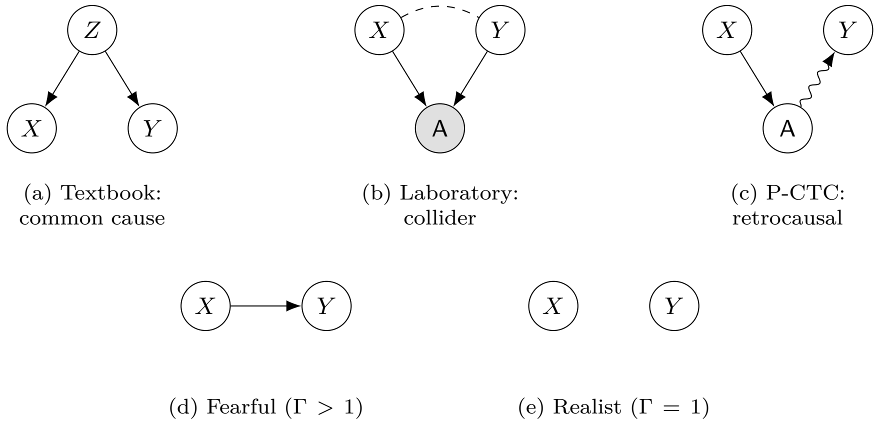
The effective normal-form entries are therefore the original payoff entries multiplied by these sector weights, as in Eq. (13).

A. Success-weighted Newcomb payoffs

To put the two roles on symmetric footing, define the second payoff by swapping arguments,

$$u_2(x, y) = u_1(y, x). \qquad (22)$$

The swapped payoff u_2 makes the match and mismatch sectors appear as a symmetric normal form; Y's role in that matrix is set by the analysis of Sec. VI. Under the per-attempt convention (13), each payoff entry of Eq. (14) is multiplied by the appropriate success probability. Using Eq. (4), the effective payoff matrix is



(a) Textbook: common cause

(b) Laboratory: collider

(c) P-CTC: retrocausal

(d) Fearful ($\Gamma > 1$)

(e) Realist ($\Gamma = 1$)

FIG. 2. Causal readings of the choice–forecast relation, in the act/record notation X, Y of the main text. Top row, three readings of the same correlation. The top row instantiates Reichenbach's principle: a correlation between X and Y is explained by a common cause (a) or a direct cause (c), the collider (b) being the excluded case in which conditioning on a common effect manufactures the association [48, 49]. (a) Textbook Newcomb routes it through a common cause Z, the agent's disposition, realized as the process matrix W of App. L. (b) The laboratory reading routes it through a collider: X and Y are prepared independently and the acceptance A is their common effect; conditioning on A (shaded), as conditional scoring does, opens the dashed induced X–Y association, the Simpson reversal of Sec. IX B, which per-attempt scoring leaves absent. The correlation is non-causal. (c) The P-CTC reading reverses the record's edge: the postselection is a final-state boundary condition, so A acts back on the record ($\mathsf{A} \to Y$, wavy, backward in time), giving the chain $X \to \mathsf{A} \to Y$ and a causal correlation. Readings (b) and (c) are time-reverses of one structure, the time-reversal invariance of Sec. III D; reversing both collider edges instead gives (a). Bottom row, the two Wolpert–Benford factorizations the collider interpolates: (d) the Fearful structure, the forecast conditioned on the act ($X \to Y$), for $\Gamma > 1$, and (e) the Realist structure, whose y-independent conditional $P(x \mid y) = h(x)$ makes X and Y independent, at $\Gamma = 1$. The verdict is reported at the symmetric prior, where the scoring conventions agree.

$$\begin{array}{c|cc} & C & D \\ \hline C & \big(p_{\text{match}}M,\ p_{\text{match}}M\big) & \big(p_{\text{mismatch}}\cdot 0,\ p_{\text{mismatch}}(M+m)\big) \\ D & \big(p_{\text{mismatch}}(M+m),\ p_{\text{mismatch}}\cdot 0\big) & \big(p_{\text{match}}m,\ p_{\text{match}}m\big) \end{array}. \tag{23}$$

Comparing with Eq. (21) identifies

$$\begin{aligned} R &= p_{\text{match}}M, & P &= p_{\text{match}}m, \\ T &= p_{\text{mismatch}}(M+m), & S &= 0. \end{aligned} \tag{24}$$

The temptation payoff $M + m$ lives off-diagonal in the Newcomb table: it is the reward for choosing D when the prior-record variable is C. In the present mechanism that payoff is carried by a mismatch profile, hence inherits the acceptance probability p_{mismatch}.

With these entries the symmetrized normal form changes ordinal type as Γ grows: it is Prisoner's-Dilemma–ordered for $\Gamma < (M+m)/M$ and Stag-Hunt/assurance–ordered for $\Gamma > (M+m)/M$, and within the coordination region (C, C) is risk-dominant for $\Gamma > (M+m)/(M-m)$. Above $\Gamma = (M+m)/m$ a strong-assurance regime sets in, $R > P > T > S$, where per attempt mutual defection outpays the unilateral temptation because the temptation branch is a suppressed mismatch. Appendix D gives the full ordinal classification with the degenerate endpoints and the proof.

These two boundaries are distinct, the second larger. In the interval between them the symmetrized table is Stag-Hunt-ordered, two diagonal fixed points exist, and the recommendation at $\beta = 1/2$ remains two-boxing because (D, D) is risk-dominant there (Theorem 1). The recommendation changes at the risk-dominance boundary, so the equilibrium and its selection arrive at separate thresholds. This interval is region II of Fig. 3.

The nontrivial condition $R > T$ reads

$$p_{\text{match}}M > p_{\text{mismatch}}(M+m), \tag{25}$$

equivalently

$$\Gamma > \frac{M+m}{M}. \tag{26}$$

When this condition holds, a second diagonal fixed point appears: (C, C), the mutual one-boxing outcome. The temptation branch remains available, but it must pass through the mismatch sector, and the postselection can make it rare per attempt. The threshold sits a distance m/M above $\Gamma = 1$, so the signal strength required to flip the ordering is the reward ratio m/M itself; for orientation, Gardner's $M =$ \$1,000,000 and $m =$ \$1,000 give $(M+m)/M \approx 1.001$.

B. Basin threshold and risk dominance

Once the Newcomb payoffs have been inserted, the condition $R > T$ marks the appearance of the one-boxing agreement equilibrium (C, C). The setup prior β of Sec. III then determines on which side of the Stag-Hunt basin boundary the preparation lies. If the agent chooses C, the per-attempt expected payoff is

$$U(C|\beta) = \beta R + (1-\beta)S.$$

If the agent chooses D, it is

$$U(D|\beta) = \beta T + (1-\beta)P.$$

The indifference prior is therefore

$$\beta_{\rm ind} = \frac{P-S}{(P-S)+(R-T)}.$$

For $\beta > \beta_{\rm ind}$, the one-boxing branch has the larger per-attempt value.

For the success-weighted Newcomb matrix, $S = 0$. Substituting Eq. (24) gives

$$\beta_{\rm ind} = \frac{P}{P+R-T} = \frac{\Gamma\, m}{(M+m)(\Gamma-1)}. \tag{27}$$

Equation (27) is the setup-prior boundary between the one-boxing and two-boxing basins. It enters the allowed interval at $\beta_{\rm ind} = 1$, which gives the Stag-Hunt boundary $\Gamma = (M+m)/M$. The one-boxing basin becomes the larger of the two at $\beta_{\rm ind} = 1/2$, which gives the risk-dominance boundary.

The latter condition is the Harsanyi–Selten risk-dominance criterion [10]. The agreement equilibrium (C, C) is risk-dominant when

$$R - T > P - S.$$

Using Eq. (24), this becomes

$$p_{\rm match}(M-m) > p_{\rm mismatch}(M+m), \tag{28}$$

equivalently

$$\Gamma > \frac{M+m}{M-m}.$$

Corollary (Unbiased setup prior)**.** *At $\beta = 1/2$, the per-attempt expected payoffs are proportional to the symmetrized normal-form payoffs. Hence risk dominance and the single-agent comparison of Theorem 1 coincide: (C, C) is risk-dominant iff one-boxing has the larger ensemble-average payoff per attempt.*

Figure 3 shows the two boundaries in the $(\theta,\, m/M)$ plane. The solid curve is the payoff-ordering boundary $R = T$. The dashed curve is the risk-dominance boundary $R - T = P - S$,

$$\theta_{\rm rd} = 2\arctan\Big[\Big(\frac{1-m/M}{1+m/M}\Big)^{1/2}\Big],$$

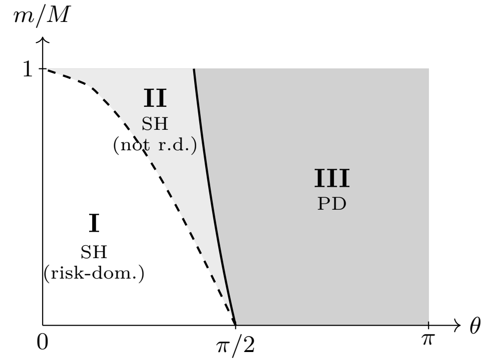


FIG. 3. Phase diagram in the $(\theta,\, m/M)$ plane, with $\Gamma = \cot^2(\theta/2)$. **I** (white): Stag-Hunt payoff ordering with one-boxing (C, C) risk-dominant. **II** (light gray): Stag-Hunt payoff ordering, but two-boxing (D, D) is risk-dominant. **III** (dark gray): Prisoner's-Dilemma ordering, where D strictly dominates. The solid curve is the payoff-ordering boundary (26); the dashed curve is the risk-dominance boundary (28). Both curves meet at $\theta = \pi/2$ in the limit $m/M \to 0$. The plotted range is $0 < m/M < 1$, corresponding to $M > m > 0$.

and it lies inside the Stag-Hunt region. As θ increases from 0, the postselection filter becomes less selective: region I has Stag-Hunt payoff ordering and risk-dominant agreement, region II has Stag-Hunt payoff ordering with (D, D) risk-dominant, and region III has Prisoner's-Dilemma ordering, with D strictly dominant. For $M \gg m$ the two curves lie close to each other near $\theta = \pi/2$, their separation set by m/M; for Gardner's values the two thresholds are $\Gamma_{\rm SH} = 1.001$ and $\Gamma_{\rm RD} \approx 1.002$. That reweighting any Prisoner's Dilemma by a strong enough diagonal-success correlation yields coordination is generic; what is specific here is that a single physical parameter $\Gamma = \cot^2(\theta/2)$ sets both thresholds and ties them to the reward ratio m/M.

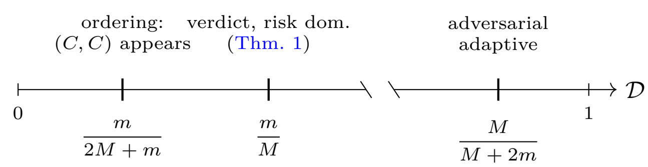


FIG. 4. The threshold ladder on the distinguishability axis $\mathcal{D} = (\Gamma-1)/(\Gamma+1)$; positions are schematic. The one-boxing equilibrium appears at $\mathcal{D} = m/(2M+m)$, the convention-independent verdict and risk dominance switch at $\mathcal{D} = m/M$, and a nonzero one-boxing Born weight survives the adversarial adaptive prior only above $\mathcal{D} = M/(M+2m)$ (Sec. VI). For Gardner's values $M = \$1{,}000{,}000$ and $m = \$1{,}000$ the three thresholds sit at $\mathcal{D} \approx 5.0 \times 10^{-4}$, $\mathcal{D} = 10^{-3}$, and $\mathcal{D} \approx 0.998$: the game-structure and verdict thresholds lie within 10^{-3} of $\mathcal{D} = 0$ and the adversarial threshold within 2×10^{-3} of $\mathcal{D} = 1$.

Figure 4 places the three thresholds on the distinguishability axis: the ordering and verdict thresholds are both of order m/M, while the adversarial adaptive threshold sits within $2m/M$ of perfect distinguishability. Section VI quantifies the separation.

Potential and basins. The symmetrized table (23) is an exact potential game whose two coordination attractors are separated by an interior repeller at β_{ind}, the deeper well switching to one-boxing at the risk-dominance boundary $\Gamma = (M{+}m)/(M{-}m)$; App. E gives the potential, barrier, and replicator dynamics. The equilibrium-selection content is the static counterpart of an attractor structure.

V. TWO NEWCOMB PROBLEMS SIDE BY SIDE

Lewis [8] reads a Prisoner's Dilemma as two Newcomb problems set side by side: each player takes the partner's correlated action as a prediction of their own, the inter-player correlation playing the role of predictor reliability. The agreement filter supplies that correlation explicitly, and the reading carries through to Stag-Hunt ordering.

Promote Y to a second decision register. Players $i = 1, 2$ choose $x_i \in \{C, D\}$; the pair is filtered by the parity channel with survival weight $w(x_1, x_2) = p_{\text{match}}$ on agreement and p_{mismatch} on disagreement; and player i takes the Newcomb payoff $u_1(x_i, x_j)$ of Eq. (14), with the partner $j \neq i$ read as the prediction of x_i. The filter enters in two equivalent ways. As a game, the per-attempt weight folds into the payoff, $U_i = w\, u_1(x_i, x_j)$; since $U_2(x_1, x_2) = w\, u_1(x_2, x_1) = w\, u_2(x_1, x_2)$, player 1's induced normal form is the symmetric table (23) with entries (24). As a parametric decision, the raw payoff u_1 is kept and the agreement bias is carried by the prediction's reliability $q = \Pr(x_j = x_i \mid \mathsf{A})$; by Eq. (11) the accepted ensemble multiplies the prior agreement odds by Γ, so at the symmetric profile $q = \Gamma/(\Gamma + 1)$ (Sec. II D). The unnormalized weight w and the conditional reliability q are the per-attempt and conditional encodings of one correlation. Write $G(\Gamma)$ for the resulting one-parameter family of games.

Read parametrically, each side of $G(\Gamma)$ is the standard Newcomb problem with payoffs u_1 and reliability $q = \Gamma/(\Gamma+1)$, so one-boxing carries the larger expected utility iff $\Gamma > (M + m)/(M - m)$. Read as a game, $G(\Gamma)$ is Prisoner's-Dilemma–ordered for $\Gamma < (M + m)/M$ and Stag-Hunt–ordered above it, with (C, C) risk-dominant iff $\Gamma > (M + m)/(M - m)$ (Prop. D). The two readings turn to one-boxing at the same $\Gamma = (M + m)/(M - m)$: $G(\Gamma)$ is two Newcomb problems side by side in Lewis's sense throughout $\Gamma \geq 1$, the consistency factor setting only the equilibrium structure of the game they compose into, and the crossover at $\Gamma = (M + m)/M$ carries that composite from Prisoner's-Dilemma to Stag-Hunt ordering. Appendix F states this as a proposition and proves it.

For $\Gamma < 1$ the same normal-form family is anti-correlated; relabeling the record returns it to the predictor regime. We include that side only when classifying the full ordering diagram.

Remark. Stag-Hunt ordering means defection ceases to dominate the composite ($R > T$) while the Newcomb conflict remains in the raw payoff, where two-boxing pays $+m$ against either prediction. The lower threshold $(M + m)/M$ has no single-agent counterpart: it marks the appearance of a second equilibrium, visible only with the two problems placed side by side. Bermúdez [31] argues against Lewis that the parametric and strategic readings come apart, the epistemic conditions that give Newcomb's problem its force being the ones that keep it from a Prisoner's Dilemma; the consistency factor renders that gap quantitative, the readings agreeing only at $\Gamma = (M + m)/(M - m)$, the unique strength at which Lewis's identification closes.

The composite Stag Hunt is a per-attempt object: conditional scoring renormalizes the weight w away and restores entrywise dominance, and at the symmetric profile the verdict is convention-independent in any case (Theorem 1), so the decomposition holds where the invariant core lives.

VI. ADAPTIVE PREPARATION: THE PREDICTOR'S OBJECTIVE

The fixed-prior comparison of Sec. III treats the preparation of Y as external. By Assumption 1, Ω knows the agent's preparation; β may therefore respond to it, and the adaptive verdict depends on what Ω optimizes. Newcomb's original predictor is closer in spirit to an accuracy-seeking forecaster than to an adversary. We therefore separate three objectives, holding the postselection mechanism, and hence Γ, fixed. The derivations are collected in Appendices G–I.

Let the decision qubit be prepared as

$$|\psi(\alpha)\rangle_X = \sqrt{\alpha}\, |C\rangle + \sqrt{1-\alpha}\, |D\rangle\,,$$

so that the final computational-basis readout gives C with Born weight α and D with Born weight $1{-}\alpha$. The realized value of X is still not read out before the postselection. Knowing α, Ω chooses the prior-record preparation

$$\rho_Y(\beta) = \beta\, |C\rangle\langle C| + (1-\beta)\, |D\rangle\langle D|\,.$$

The agent's decision is the preparation α, a probability of one-boxing; the act X is its downstream Born-rule readout. A relative phase in the preparation of X leaves the calculation unchanged, because the postselection effect, the prior-record state, and the payoff observable are all diagonal in the $\{C, D\}$ basis.

For a general symmetric 2×2 normal form with entries (R, S, T, P), the per-attempt ensemble payoff is

$$\begin{aligned} W(\alpha, \beta) = {}& \alpha\beta R + \alpha(1-\beta) S \\ &+ (1-\alpha)\beta T + (1-\alpha)(1-\beta) P. \end{aligned} \tag{29}$$

Equivalently,

$$\begin{aligned} W(\alpha, \beta) &= P + \alpha(S - P) + \beta(T - P) + \alpha\beta\Delta, \\ \Delta &= R - S - T + P. \end{aligned} \tag{30}$$

Thus

$$\frac{\partial W}{\partial \alpha} = S - P + \beta\Delta, \qquad \frac{\partial W}{\partial \beta} = T - P + \alpha\Delta.$$

For the Newcomb entries (24), $\Delta = (M+m)(p_{\text{match}} - p_{\text{mismatch}})$, so $\Delta > 0$ for $\Gamma > 1$ and $\Delta = 0$ at $\Gamma = 1$.

The three adaptive objectives give sharply different thresholds.

Forecasting predictor. If Ω maximizes forecast accuracy, its best response is the threshold rule

$$\beta_{\text{MAP}}(\alpha) = \mathbf{1}[\alpha > \tfrac{1}{2}], \tag{31}$$

with the tie at $\alpha = \frac{1}{2}$ assigned to $\beta = 0$. Optimizing against this response gives

$$\alpha^*_{\text{MAP}} = 1 \qquad \text{for all } \Gamma > \frac{M+m}{M}. \tag{32}$$

Thus an accuracy-seeking predictor makes pure one-boxing optimal throughout the Stag-Hunt region. In the Prisoner's-Dilemma region the value is only a supremum as $\alpha \to \frac{1}{2}^+$: the agent is forecast as a one-boxer while keeping as much two-boxing weight as possible (Appendix G).

Calibrated predictor. If Ω sets $\beta = \alpha$, then $W(\alpha, \alpha)$ is convex in α for $\Gamma > 1$, so the optimum is at a corner. Since $R > P$,

$$\alpha^*_{\text{cal}} = 1 \qquad \text{for all } \Gamma \geq 1. \tag{33}$$

The same verdict holds for a self-calibrated predictor satisfying $\beta = \Pr(X = C \mid \mathsf{A})$ (Appendix H). Functional decision theory recommends one-boxing [6]; the calibrated column returns that recommendation as an output of the postselection mechanism, one-boxing being selected independently of the mechanism's threshold structure whenever the predictor is calibrated.

Adversarial predictor. If Ω selects β to minimize the agent's payoff, the robust adaptive value is the lower envelope

$$V_{\text{adv}}(\alpha) := \inf_{\beta \in [0,1]} W(\alpha, \beta).$$

The robust optimum is

$$\alpha^*_{\text{adv}} = \begin{cases} 0, & \Gamma \leq \dfrac{M+m}{m}, \\ \dfrac{\Gamma m - (M+m)}{(M+m)(\Gamma - 1)}, & \Gamma > \dfrac{M+m}{m}. \end{cases} \tag{34}$$

The corresponding dual prior in the mixed regime is

$$\beta^* = \frac{\Gamma m}{(M+m)(\Gamma - 1)}, \qquad \beta^* - \alpha^* = \frac{1}{\Gamma - 1}. \tag{35}$$

As $\Gamma \to \infty$, both α^* and β^* tend to $m/(M+m)$. Even perfect consistency therefore leaves the lower-envelope preparation overwhelmingly two-boxing in the standard Newcomb regime $M \gg m$. For Gardner's values $M = 10^6$, $m = 10^3$, the adversarial threshold is 1001, and the ideal-limit one-boxing weight is 9.99×10^{-4} (Appendix I).

No additional P-CTC assumption is introduced here: the postselected map contributes only the sector survival weights, and adaptive preparation changes the input ensemble fed to the same trace-decreasing map. With noise, the formulas hold with Γ replaced by the effective consistency factor Γ_{eff} (Sec. VIII).

Objective	response $\beta_O(\alpha)$	opt. α	onset
MAP forecast	$\mathbf{1}[\alpha > \frac{1}{2}]$	1	C for $\Gamma > \dfrac{M+m}{M}$
Calibrated	α	1	C for all $\Gamma \geq 1$
Adversarial	$\arg\min_\beta W$	$0 \to \dfrac{m}{M+m}$	nonzero C-weight for $\Gamma > \dfrac{M+m}{m}$

TABLE II. Optimal decision-qubit preparation under three predictor objectives, same postselection filter throughout. The accuracy-seeking predictor is the Newcomb-faithful objective; the adversarial column is a robustness bound.

VII. THE PREDICTOR'S PREDICAMENT: A DISTINGUISHABILITY–VISIBILITY COMPLEMENTARITY

Once the act is the computational-basis readout of a decision qubit, that limitation is bounded by a complementarity relation: the predictor's accuracy (the distinguishability $\mathcal{D}$) and the act register's visibility are complementary quantities, constrained by an inequality the coherent implementation meets with equality. This locates the construction's quantum content in the predictor's inability to acquire the act as an early classical fact, while the decision itself stays classical (Sec. III).

Keep the decision qubit $|\psi(\alpha)\rangle_X = \sqrt{\alpha}\,|C\rangle + \sqrt{1-\alpha}\,|D\rangle$ of Sec. VI, with the act the outcome $x \in \{C, D\}$ of the later readout. For this section widen the predictor to its broadest form: any process that, before the final act readout, couples X to a record system Y, whose computational-basis readout y is the predictor's forecast. Let

$$q := \Pr(y = x) \tag{36}$$

be its reliability in the sense of Sec. II D, now for an arbitrary record y, $\mathcal{D} := 2q - 1 \in [0, 1]$ the corresponding *distinguishability*, extending Eq. (12) to that record ($\mathcal{D}$, distinct from the action label D), and

$$\mathcal{V} := \sqrt{\langle \sigma_x \rangle_X^2 + \langle \sigma_y \rangle_X^2}$$

the phase-optimized transverse visibility of the act register, its surviving coherence in the plane complementary to the payoff basis. Equivalently, if $\rho_{CD} := \langle C|\rho_X|D\rangle$, then $\mathcal{V} = 2|\rho_{CD}|$: the "fringe" is the phase response of the two-state act register, obtained by inserting a relative phase between C and D and reading in the complementary basis. In the real-phase $R_y(\theta)$ implementation used below, this visibility lies in the σ_x quadrature, so $\mathcal{V} = |\langle\sigma_x\rangle_X|$. For the parity-filter family, the predictor regime is $q \geq 1/2$, equivalently $\Gamma \geq 1$ ($\theta \in [0, \pi/2]$); for $\Gamma < 1$ the record is anti-correlated with the act, and $\mathcal{D}$ is taken after optimal relabeling. Fig. 3 shows $\Gamma < 1$ only to complete the ordering classification.

Proposition (Distinguishability–visibility complementarity)**.** Let X be the act qubit and Y a record system with a designated computational-basis readout y, with $q = \Pr(y = x)$, distinguishability $\mathcal{D} := 2q - 1 \in [0, 1]$, and phase-optimized transverse visibility $\mathcal{V}$. Then

$$\mathcal{D}^2 + \mathcal{V}^2 \leq 1, \tag{37}$$

with equality in the pure optimal-record case used below. *Proof.* A which-path argument: the bound is Englert's relation [28, 29] for the record Y monitoring the act X, with equality in the pure optimal-record case. See App. J. □

The parity filter realizes the bound. At the symmetric preparation $\alpha = \beta = 1/2$, the coherent filter $K = c\,\Pi_= + e^{i\varphi} s\,\Pi_{\neq}$ has accepted-ensemble distinguishability $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$ and visibility $\mathcal{V} \leq 2\sqrt{\Gamma}/(\Gamma + 1)$. The real-phase circuit of Fig. 1 places this visibility entirely in the σ_x quadrature and meets the bound with equality,

$$\mathcal{D} = \frac{\Gamma - 1}{\Gamma + 1} = \cos\theta, \qquad \mathcal{V} = \sin\theta, \qquad \mathcal{D}^2 + \mathcal{V}^2 = 1. \tag{38}$$

The accepted state $|\psi(\theta)\rangle$ (App. J) is pure, and $\rho_X = \frac{1}{2}(I - \sin\theta\,\sigma_x)$ gives $\mathcal{V} = \sin\theta$; $q = \Gamma/(\Gamma + 1)$ (Sec. II D) gives $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1) = \cos\theta$ via $\Gamma = \cot^2(\theta/2)$. At the symmetric preparation, the two conditional states of Y associated with $X = C$ and $X = D$ are $c\,|0\rangle - s\,|1\rangle$ and $-s\,|0\rangle + c\,|1\rangle$; their density-matrix difference is $\cos\theta\,\sigma_z$, so the computational-basis readout of Y is an optimal which-way measurement. The record is therefore optimal and the bound is met with equality. A dephased implementation with the same sector survival weights gives $\mathcal{V} = 0$; the sector survival weights fix $\mathcal{D}$ but leave the transverse coherence implementation-dependent.

Corollary (No early classical fact)**.** For a genuine superposition $0 < \alpha < 1$, any record that predicts the act with certainty ($q = 1$, equivalently $\Gamma \to \infty$) forces $\mathcal{V} = 0$: the act register is dephased to $\rho_X = \alpha|C\rangle\langle C| + (1-\alpha)|D\rangle\langle D|$, with no coherence in the complementary basis. Perfect prediction therefore operationally coincides with dephasing in the act basis: the record fixes the later act readout only by removing the complementary coherence. A record that kept that coherence would carry a $\sigma_z \otimes \sigma_x$ cross-register term, the causally-nonseparable case of App. L; the diagonal filter never produces it. The record–act consistency the predictor requires can be imposed only as a joint constraint on (X, Y), which is what the postselected map $K\rho K^\dagger$ does.

Proof. With $\Pr(y = x) = 1$ the joint readout is supported on the diagonal, so tracing out Y leaves ρ_X diagonal and $\mathcal{V} = 0$; the endpoint $\Gamma \to \infty$ of (38) realizes it. See App. J. □

Γ fixes $\mathcal{D}$ and leaves $\mathcal{V}$ free: the same sector survival weights admit a coherent implementation, $\mathcal{V} = \sqrt{1 - \mathcal{D}^2}$, and a dephased one, $\mathcal{V} = 0$, and the visibility distinguishes the two. The verdict of Eq. (19) is carried by $\mathcal{D}$ alone; the visibility is its complementary partner. It is decision-orthogonal, registering whether the consistency condition was realized coherently or by a classical reliability oracle. It is maximal at $\Gamma = 1$, where $\mathcal{D} = 0$, and zero at the corner $(\mathcal{D}, \mathcal{V}) = (1, 0)$, the perfect-prediction limit $\Gamma \to \infty$, the Omega point at which Ω never errs. Figure 5 summarizes the geometry.

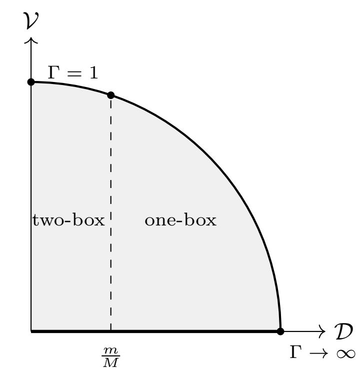


FIG. 5. Decision content and quantum content in the $(\mathcal{D}, \mathcal{V})$ plane at the symmetric preparation. Physical records lie in the quarter disk $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$ (shaded, Eq. (37)). The coherent implementation traces the unit arc $\mathcal{V} = \sqrt{1 - \mathcal{D}^2}$, from $\Gamma = 1$ to the Omega point at $(1, 0)$, the ideal P-CTC limit; a dephased implementation lies on the $\mathcal{V} = 0$ axis (thick segment). The verdict of Eq. (19) reads only the horizontal coordinate: one-box for $\mathcal{D} > m/M$ (dashed line drawn schematically; for $M \gg m$ the true threshold m/M is close to $\mathcal{D} = 0$ and one-boxing occupies almost the whole axis, as in Fig. 4). The visibility is maximal at $\Gamma = 1$ and vanishes in the perfect-prediction limit.

A. The σ_x signature at the symmetric preparation

The basis-diagonal decision content of Secs. III–VI is classical: a rejection sampler at matching sector survival weights reproduces all expected utilities. The construction does, however, leave an experimentally accessible signature whose visibility, by the bound (37), is complementary to the decision variable itself. For the real-phase $R_y(\theta)$ realization, the visibility lies in the σ_x quadrature, and the expectation on X at $\alpha = \beta = 1/2$ is $\langle\sigma_x\rangle_X = -2\sqrt{\Gamma}/(\Gamma + 1)$, while the dephased classical model gives zero (Table III).

Anything that distinguishes the construction from this classical model must live off-diagonal in $\{C, D\}$; the uniqueness proposition of Sec. II E covers basis-diagonal observables and does not extend to σ_x, which is why it can separate the coherent and dephased implementations. At the symmetric preparation $\alpha = \beta = 1/2$ both registers are in σ_x eigenstates $|+\rangle = (|C\rangle + |D\rangle)/\sqrt{2}$. Sec. II noted that $R_y(\theta)$ on the ancilla supplies the non-triviality at the postselection; the symmetric preparation places the registers themselves on σ_x eigenstates as well, and the σ_x measurement on X reads off what the construction does with this coherent input.

Carrying the $|+\rangle_X |+\rangle_Y$ input through the two CNOTs, the ancilla rotation, and postselection on $|0\rangle_a$ gives a pure joint state $|\psi(\theta)\rangle_{XY}$ (App. J), with acceptance probability 1/2 independent of θ and accepted match/mismatch weights $\cos^2(\theta/2), \sin^2(\theta/2)$ that recover $\Gamma = \cot^2(\theta/2)$. Its reduced state on X is $\rho_X(\theta) = \frac{1}{2}(I - \sin\theta\, \sigma_x)$, so $\langle\sigma_x\rangle_X = -2\sqrt{\Gamma}/(\Gamma + 1)$. The sign is negative because agreement-favoring postselection retains the branch whose Kraus amplitude on the mismatch sector carries the relative minus sign of Eq. (7), imprinting an anti-aligned σ_x coherence. The classical rejection-sampling model gives $\rho_X^{\rm cl} = I/2$ and $\langle\sigma_x\rangle_X^{\rm cl} = 0$ at every Γ, so a single observable separates the coherent construction from any classical rejection sampler with the same computational-basis sector survival weights.

The sector survival weights fix every basis-diagonal payoff comparison; the off-diagonal sector carries the remaining quantum content, whose local act-register signature depends on the implementation phase and is maximal for the real-phase $R_y(\theta)$ circuit used here (App. J).

The σ_x signature is strongest at $\Gamma = 1$, where $\langle\sigma_x\rangle_X = -1$: at the indifference point neither sector is reweighted, the filter maps $|+\rangle |+\rangle$ to $|-\rangle |-\rangle$, and the coherence is carried through at full strength with its sign reversed, a maximum at the point where the distinguishability vanishes ($\mathcal{D} = 0$). At the endpoints the joint state stays Bell-coherent ($|\Phi^+\rangle$ as $\Gamma \to \infty$, $|\Psi^+\rangle$ as $\Gamma \to 0$) while the reduced state on X becomes $I/2$ and the local signature vanishes. By exchange symmetry $\langle\sigma_x\rangle_Y = \langle\sigma_x\rangle_X$.

The symmetric preparation has decision-theoretic significance: at $\beta = 1/2$ the per-attempt and conditional conventions agree (Theorem 1), and risk dominance coincides with the single-agent expected-payoff comparison (Sec. IV B). The σ_x signature is off-strategic: pure two-boxing ($\alpha = 0$) and pure one-boxing ($\alpha = 1$) leave the registers in computational-basis states with $\langle\sigma_x\rangle = 0$, so it lies in the complement of the payoff basis and leaves the strategic verdict untouched (Eq. (37)). The coherence maximum at $\Gamma = 1$ sits in the PD-ordered region where the verdict is two-boxing, and its vanishing at the ideal-P-CTC limit is where Nozick's perfect-predictor stipulation is realized.

Operationally, the discriminator runs the same survival-weight statistics twice, once with coherent $|+\rangle_X |+\rangle_Y$ preparation and once with a dephased classical mixture matched on the computational-basis statistics: the basis-diagonal payoffs agree, while only the coherent run carries the predicted σ_x expectation. A nonzero measurement witnesses off-diagonal coherence in the postselected map, a device-certification question distinct from the decision verdict, confirming that the postselection was carried out coherently rather than emulated by a classical reliability oracle with matching survival weights. Under the noisy-channel reading of Sec. VIII the averaged numerator may carry cross-sector coherences that $\Gamma_{\rm eff}$ does not capture, so the discriminator requires sharp control of θ. Appendix J gives the linear-optical implementation and the sampling cost.

quantity at $\alpha = \beta = \frac{1}{2}$	coherent $R_y(\theta)$	dephased
heralding $\Pr(\mathsf{A})$	1/2	1/2
accepted agreement q	$\Gamma/(\Gamma + 1)$	$\Gamma/(\Gamma + 1)$
visibility $\mathcal{V}$	$\sin\theta$	0
$\langle\sigma_x\rangle_X$	$-2\sqrt{\Gamma}/(\Gamma + 1)$	0

TABLE III. Coherent parity filter against a dephased model matched on the computational-basis sector survival weights, at the symmetric preparation. The basis-diagonal entries ($\Pr(\mathsf{A})$, q) agree by construction; the transverse coherence ($\mathcal{V}$, $\langle\sigma_x\rangle_X$) separates the two. The contrast is largest at $\Gamma = 1$, where $\langle\sigma_x\rangle_X = -1$ against 0.

VIII. NOISE AND THE EFFECTIVE CONSISTENCY FACTOR

In the time-travel reading, noise in the P-CTC channel is imperfect control of the consistency angle θ. At $\theta = 0$ mismatch histories are eliminated; finite $\Gamma = \cot^2(\theta/2)$ lets them survive at relative rate Γ^{-1}. A run-to-run distribution of θ averages to the same two-sector form with survival ratio $\Gamma_{\rm eff}$. For basis-diagonal payoff comparisons with unobserved θ, only the averaged agreement and disagreement survival weights enter. What the per-run θ would add is taken up at the end of this section.

Averaging the accepted-branch numerator $K(\theta)(\cdot)K^\dagger(\theta)$ over the angle noise yields an effect of the same two-sector form as Eq. (9), because $E(\theta) = K^\dagger K$ is diagonal in the sector basis for each θ and diagonal matrices average to diagonal matrices:

$$E_{\rm eff} = \mathbb{E}\big[\cos^2(\theta/2)\big]\,\Pi_= + \mathbb{E}\big[\sin^2(\theta/2)\big]\,\Pi_{\neq}. \tag{39}$$

The corresponding effective consistency factor is

$$\Gamma_{\rm eff} = \frac{\mathbb{E}\big[\cos^2(\theta/2)\big]}{\mathbb{E}\big[\sin^2(\theta/2)\big]}. \tag{40}$$

Averaging over the angle noise places the single-θ numerator of Sec. II in the noisy-P-CTC framework of Ji et al. [19] with an explicit channel. Their noisy P-CTC is a noiseless P-CTC followed by a channel $\mathcal{N}$, entering their map through its Choi state $\mathcal{N}[\Phi]$. For the basis-diagonal decision content the averaged filter is this construction with $\mathcal{N}$ the bit-flip channel

$$\mathcal{N}_\ell(\sigma) = (1 - \ell)\,\sigma + \ell\,\sigma_x \sigma \sigma_x, \qquad \ell = \mathbb{E}[\sin^2(\theta/2)], \tag{41}$$

whose flip rate is the mean mismatch weight ℓ: agreement survives at $1-\ell$, disagreement at ℓ, and the survival ratio is $\Gamma_{\text{eff}} = (1-\ell)/\ell$, Eq. (40). Every basis-diagonal Newcomb threshold of Secs. IV–VI then follows from the sharp case by replacing Γ with Γ_{eff}: the ideal agreement-projector limit is $\Gamma_{\text{eff}} \to \infty$ ($\ell \to 0$), and $\Gamma_{\text{eff}} = 1$ ($\ell = \frac{1}{2}$) carries no match-mismatch consistency signal.

The boundaries are mismatch tolerances in ℓ: Stag-Hunt ordering holds for $\ell < M/(2M+m)$, risk dominance of the agreement equilibrium for $\ell < (M-m)/(2M)$, and the adaptive-preparation verdict for $\ell < m/(M+2m)$. For $M \gg m$ the first two tolerances are close to 1/2 while the adaptive one is of order m/M, so the adaptive condition is what tests whether the consistency signal stays sharp enough to move the agent's preparation.

Since θ is a rotation angle, its noise wraps around the circle, where the von Mises distribution of concentration κ plays the role a Gaussian does on the line. It gives the mismatch weight in closed form:

$$\ell = \mathbb{E}[\sin^2(\theta/2)] = \tfrac{1}{2}\big(1 - A(\kappa)\big), \qquad A(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)}, \tag{42}$$

with A the mean resultant length and I_ν the modified Bessel functions. The adaptive verdict survives while $\ell < m/(M+2m)$, that is $A(\kappa) > 1 - 2m/(M+2m)$. At Gardner's values $M = 10^6$, $m = 10^3$ this is $\kappa \gtrsim 251$. In the concentrated limit von Mises approaches a wrapped Gaussian of variance $\sigma^2 = 1/\kappa$, where $\ell \simeq \sigma^2/4$ and the threshold reads $\sigma < 2\sqrt{m/(M+2m)} \sim 2\sqrt{m/M}$, about 0.063 rad or 3.6°. The concentration is the operational sharpness of the consistency condition, the control precision of the optical rotation in the heralded reading; a convincing experiment would keep it well above this bound.

The opposite extreme, θ uniform on $[0,\pi]$, is the maximal-ignorance limit. Run by run the realized ordering splits close to evenly: Stag-Hunt with probability $\Pr(\text{SH}) = \theta^*/\pi$, $\theta^* = 2\arctan\sqrt{M/(M+m)}$, leaving a small Prisoner's-Dilemma excess $\Pr(\text{PD}) - \Pr(\text{SH}) = m/(\pi M) + O(m^2/M^2)$ set by the reward ratio. The averaged game is nonetheless Prisoner's-Dilemma-ordered, $\Gamma_{\text{eff}} = 1$: the payoff ordering is nonlinear in the sector weights, so the ordering of the average departs from the run-by-run majority (App. K).

The averaging that produces Γ_{eff} is a map from a quantum process to its classical core. Dephasing the registers removes the σ_x coherence of Sec. VII A and leaves the diagonal survival weights of the classical rejection sampler (Sec. III D); angle noise then moves Γ to Γ_{eff} within that core. In the Aráujo–Feix–Baumeler–Wolf process the same quantum-to-classical step keeps the causal-inequality violation, which sits on the diagonal [47]; here the diagonal carries the decision content of Theorem 1, classical and causally separable (Sec. IX), and decoherence removes only the decision-orthogonal coherence. Coherence and causal order are independent axes, and the von Mises concentration κ of Eq. (42) sets a closed-form path along the first.

IX. DISCUSSION

The postselected update is a joint conditioning followed by renormalization (Eq. (2)), which preserves correlations while enforcing a consistency bias. In the present construction, that bias is a parity-selective consistency condition controlled by the rotation angle θ. As Γ is varied one can read the change either as a shift in predictor reliability within the accepted ensemble or as a shift in the payoff ordering of the induced normal form. Both effects are consequences of how the mechanism reweights match and mismatch histories. Which reading governs depends on whether one weights histories by their survival probability (per attempt) or conditions on acceptance. The analysis inherits the interpretive caveats of the P-CTC framework, made precise below.

A. Operational and P-CTC readings of the consistency condition

The construction supports three claims of decreasing strength, and we make only the first two. (i) *Operational*: a laboratory postselected parity filter has the stated sector survival weights $p_{\text{match}}, p_{\text{mismatch}}$, whatever their physical origin. (ii) *Structural*: this trace-decreasing CP numerator followed by renormalization has the postselected-map form of the P-CTC framework [14], realized as a final-state boundary condition on the ancilla; the payoff-ordering and coherence results use only this CP numerator and its sector survival weights. (iii) *Physical*: that the filter realizes a genuine closed timelike curve in spacetime. We assert (i) and (ii) and bracket (iii): the results neither require an actual CTC nor rule one out; whether the cyclic reading is laboratory feedback or a genuine CTC depends on how the map is embedded in spacetime [23], which we leave to future work. The Newcomb content of the results rests on (i) and (ii) alone, for the reason given below.

The structural reading connects to black-hole final-state proposals [35]: the trace-decreasing numerator followed by renormalization is a final-state boundary condition, with Γ setting its strength, maximally mixed at $\Gamma = 1$ and a rank-reduced projector as $\Gamma \to \infty$. Ji et al. [19] limit retrocausal communication through such maps under arbitrary final-state boundary conditions, these models included, whose unitarity Lloyd and Preskill examine [37] and whose interactions with infalling matter can break it [36].

Promoting the heralded filter to a genuine spacetime resource is the open question of reading (iii). A geometric substrate would be a Lorentzian metric with closed timelike curves (the Gödel universe, traversable wormholes [40], Gott's cosmic strings); whether such a metric can form is the content of the chronology protection conjecture [38], which holds that semiclassical backreaction forbids closed timelike curves. The filter retains only histories in which Y agrees with X; Novikov self-consistency imposes the same requirement on fields around such curves [39]. The measurement and decision results hold for the laboratory circuit whatever the embedding.

For basis-diagonal payoff data with unobserved θ, the accepted ensemble gives the averaged survival ratio Γ_{eff} (Sec. VIII). The ordering transition comes from the consistency condition changing the throughput of the branches that carry the relevant payoffs. As in Sec. II, $R_y(\theta)$ acts on the postselection basis, which is what lets the parity loop stay normalizable where the grandfather circuit has vanishing amplitude for mismatch histories.

Two features of postselected CTCs bear on the decision reading. First, the normalized P-CTC update is nonlinear in the input state, while the per-attempt quantity $\mathrm{Tr}[U\,K\rho K^\dagger]$ is linear in the unnormalized postselection numerator but is not a conditional expectation over accepted runs; the per-attempt comparison is therefore among preparations. Second, postselection confers the computational power of PostBQP = PP [22]; the open question is whether the postselected consistency condition can be realized as a spacetime resource or only as a heralded laboratory operation [21]. Because acceptance is heralded classically, the σ_x discriminator carries no superluminal signaling, and the state-discrimination power of Deutsch-type closed timelike curves [41] is absent: the postselected family is the weaker model [42], its correlations confined to the heralded ensemble.

Process-matrix membership turns on the normalization. The normalized map is linear only when the acceptance probability is preparation-independent, and that linear case is what Aráujo, Guérin and Baumeler identify as a process matrix, the tame regime of a P-CTC bounded by $\mathrm{BQP}_{\ell\mathrm{CTC}} \subseteq \mathrm{PP}$ [45]. By Theorem 1(i) this holds at $\beta = 1/2$, where $\partial \Pr(\mathsf{A})/\partial\alpha = 0$: acceptance invariance, agreement of the two scoring conventions, and membership in the process-matrix class are one condition, and the Simpson reversal at $\beta \neq 1/2$ marks the nonlinear regime outside it. That process matrix is causally separable, the parity filter being classical and diagonal. Appendix L gives it explicitly with the validity check, and locates it against the $\sigma_z \otimes \sigma_x$ bipartite [46] and three-party classical [47] processes that do violate causal inequalities.

One objection should be met directly. In the laboratory reading the filter does not forecast: Y becomes correlated with X through joint postselection, with no agent committed before the choice is made, so the literal "predictor has already acted" gloss is absent. In the reading that supplies that gloss, Ω as a time traveler enforcing consistency along a closed timelike curve, realizability is the open question just named. The two readings appear to trap the construction: Newcomb without realizability, or realizability without Newcomb. The trap dissolves once Newcomb's problem is identified with its probabilistic structure. Wolpert and Benford prove the paradox time-reversal invariant: it is unchanged whether Ω forecasts before the choice or observes it after (§3.4 of [9]). Temporal priority carries none of the problem's force, which lies in the structure relating record and act; the parity filter supplies it. The measurement and decision results of Secs. II–VII hold unconditionally for the laboratory circuit, and by time-reversal invariance they are Newcomb's problem in this structural sense. The closed-timelike-curve reading is an additional interpretation that recovers the temporal gloss as well; the gloss alone is conditional on realizability.

A further interpretive question concerns what it means for an agent to occupy a postselected branch. Per-attempt scoring treats the agent as the designer of a preparation, evaluated before the filter acts; conditional scoring treats them as the occupant of an accepted history. The two stances evaluate utility over different reference classes, and the construction keeps both available. Theorem 1 shows that the Newcomb verdict at the unbiased prior is insensitive to the stance, which is what permits a physical reading of the problem without first settling the decision-theoretic status of postselected agency.

B. Simpson's reversal and the postselection flag

The coexistence of a per-attempt one-boxing advantage and conditional-on-success two-boxing dominance is a Simpson reversal [20]. The acceptance event A depends on both x and y, so conditioning on A is collider conditioning, and the size of the reversal is set by the prior. The per-attempt and conditional conventions run parallel to the interventionist and evidential treatments of Newcomb at the level of which probabilities weight the payoffs. The postselection collider is a common effect of X and Y, while the textbook setup turns on a common cause (Fig. 2), so we keep the per-attempt and conditional comparisons separate from the causal semantics of the CDT/EDT pairing. In Reichenbach's terms [48], the textbook common cause (a) and the P-CTC direct cause (c) are the two causal explanations of the correlation, and the collider (b) is the case the principle excludes; the CDT/EDT pairing, the per-attempt and conditional conventions, and the reversal itself are one disagreement over which structure governs.

At $\beta = 1/2$ there is no reversal. The acceptance-rate gradient is $(p_{\text{match}} - p_{\text{mismatch}})(2\beta - 1)$, so by Theorem 1(i) the preparation cannot move the acceptance rate there, and the two conventions agree in sign. Away from $\beta = 1/2$ the preparation does move the acceptance rate, and the success event couples X and Y through the survival weights. Conditional scoring then splits by level. Entrywise, for each fixed y among accepted runs, it removes the survival factor and restores the dominance $u(D, y) - u(C, y) = m$, independent of θ. At the policy level it integrates over Y against the posterior $\Pr(Y \mid \mathsf{A}, X = x)$, which depends on the agent's policy through the same coupling, and the verdicts separate. The separation is widest as $\beta \to 1$: conditional scoring approaches the entrywise two-boxing comparison of M against $M + m$, while per-attempt scoring favors one-boxing across the Stag-Hunt region. The unbiased setup is therefore the symmetric reference case for the Nozick-style stipulation; an asymmetric prior adds a base-rate bias the original statement does not contain, and the reversal records its effect.

C. Related work

The construction relates to the established quantum game theory program. Beginning with Meyer [24] on quantum penny flip and Eisert, Wilkens and Lewenstein [25] on the quantum Prisoner's Dilemma, and extended by Piotrowski and Sładkowski [27] and others into economics, that literature situates quantum effects in the strategy space: classical mixed strategies are replaced with unitary operations on qubits, and new equilibria emerge from the entanglement structure of the initial state, with the Prisoner's Dilemma rendered trivial in the limit of full entanglement. Piotrowski and Sładkowski [15] and Chi and Jeong [16] treat Newcomb's problem itself in this framework, again enlarging the strategy sets to unitary operations. Cavalcanti [17] draws a different quantum connection, reading a Bell scenario as a Newcomb analogue with implications for causal decision theory. The present construction sits at a different point in that design space: strategies remain classical, the agent's binary choice of preparation, the quantum effect enters through the postselection on the parity sector, and the game-theoretic consequence is a transition from PD to Stag-Hunt ordering at the single threshold $\Gamma = 1 + m/M$ with the strategy class held fixed.

The accepted-ensemble statistics are those of a pre- and post-selected ensemble [43], and a weak measurement of σ_x before the postselection reads the corresponding weak value [44].

The symmetric game points to a dynamical extension, developed in a companion paper: attractor dynamics in the presence of time machines. The coordination region holds two self-consistent histories, the agreement profiles (C, C) and (D, D), realizing in Newcomb form the multiplicity of self-consistent CTC solutions [34]; in the landscape of Sec. IV B they are its two basins. The deeper well switches at the risk-dominance boundary $\Gamma = (M + m)/(M - m)$ of Sec. IV; above it (C, C) is risk-dominant, below it (D, D) is. Noise in the parity filter drags Γ_{eff} down, and below the ordering boundary $\Gamma = (M + m)/M$ the one-boxing basin vanishes.

Which history is realized depends on how the multiplicity is resolved. At fixed preparations, an ideal P-CTC removes the mismatch branches, leaving unnormalized sector weights $\alpha_X\alpha_Y$ and $(1 - \alpha_X)(1 - \alpha_Y)$ for the two self-consistent histories; after renormalization these give the corresponding Born probabilities. For matched preparations, $\alpha_X = \alpha_Y$, a SWAP of the two worldline registers leaves the joint preparation invariant; this is the point of contact with the single-loop reading of the film.

X. CONCLUSION

Newcomb's predictor and the grandfather paradox are two instances of one question: how a physical theory represents a consistency constraint between a record and an event. The same postselected consistency condition realizes both, deforming from a grandfather paradox resolution [42] to the present setup which retains some inconsistency. Replacing the stipulation that Ω is highly accurate with the parity filter turns the paradox into a parameterized physical model whose strength is the survival ratio Γ, and the time-travel reading of Ω becomes the limit $\Gamma \to \infty$ of a one-parameter family, the ideal P-CTC at the end of a dial.

Three results follow. First, prediction carries the structure of a measurement: the distinguishability $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$ and the act-register visibility obey $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$, with equality for the coherent implementation, and the perfect-predictor limit operationally coincides with an act-basis measurement. Second, at the unbiased setup prior the one-boxing verdict is convention-independent and classical, optimal once $\mathcal{D}$ exceeds m/M, while the σ_x visibility witnesses coherent enforcement of the consistency condition and separates the construction from a classical reliability oracle with matching survival probabilities. Third, per-attempt scoring places the verdict inside a game whose normal form passes from Prisoner's-Dilemma to Stag-Hunt ordering at $\Gamma = (M + m)/M$, the one-boxing equilibrium risk-dominant from the verdict threshold $\Gamma = (M + m)/(M - m)$ onward; read side by side the symmetric game is two Newcomb problems, so the parametric and strategic readings agree at one strength and Lewis's identification becomes the weak-predictor end of a family. The adaptive verdict follows Ω's objective, and noise replaces Γ by Γ_{eff}.

ACKNOWLEDGMENTS

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Appendix A: Survival-weight reduction

This appendix proves the proposition of Sec. II E. By diagonality, K acts as scalar multiplication on each computational basis state. Joint bit-flip exchanges $|00\rangle \leftrightarrow |11\rangle$ and $|01\rangle \leftrightarrow |10\rangle$, forcing $K_{00} = K_{11}$ and $K_{01} = K_{10}$. Setting $k_= := K_{00} = K_{11}$ and $k_{\neq} := K_{01} = K_{10}$ gives

$$K = k_= \Pi_= + k_{\neq} \Pi_{\neq}.$$

The effect $E = K^\dagger K$ is diagonal in the same basis with diagonal entries $|K_{xy}|^2$. For basis-diagonal input

$$\rho(\alpha,\beta) = \sum_{xy} \rho_{xy}(\alpha,\beta)\, |xy\rangle\, \langle xy|$$

and basis-diagonal payoff

$$U = \sum_{xy} u(x,y)\, |xy\rangle\, \langle xy|\,,$$

the per-attempt expectation is

$$\langle U\rangle_{\text{per-att}}(\alpha) = \sum_{xy} \rho_{xy}(\alpha,\beta)\, p_{xy}\, u(x,y).$$

The policy difference between two preparations is linear in $(p_{\text{match}}, p_{\text{mismatch}})$ and homogeneous of degree one, so its sign depends only on the ratio $\Gamma = p_{\text{match}}/p_{\text{mismatch}}$.

Appendix B: Proof of Theorem 1

This appendix proves Theorem 1. Part (i) is the displayed differentiation. For (ii), at $\beta = 1/2$ the per-attempt payoffs are $\bar{u}(C) = \frac{1}{2}p_{\text{match}}M$ and $\bar{u}(D) = \frac{1}{2}[p_{\text{mismatch}}(M+m) + p_{\text{match}}m]$, so $\bar{u}(C) > \bar{u}(D)$ iff $p_{\text{match}}(M-m) > p_{\text{mismatch}}(M+m)$ iff $\Gamma > (M+m)/(M-m)$. The conditional comparison is Eq. (C5), $U_{\mathsf{A}}(D) - U_{\mathsf{A}}(C) = [(M+m) - \Gamma(M-m)]/(\Gamma+1)$, which is negative on the same set. Part (iii) substitutes $q = \Gamma/(\Gamma+1)$ into the threshold (Sec. III D). ☐

Appendix C: Utilities conditioned on success

An alternative convention evaluates utilities conditioned on postselection success: one conditions on A and then computes expected payoff. In that convention, the factor p_{xy} in Eq. (13) is renormalized away. The main text keeps p_{xy} visible by working per attempt. Concretely, let $\mathbb{P}(x,y)$ denote the joint distribution over profiles before the ancilla is measured. Conditioning on A produces a new distribution over the accepted runs,

$$\mathbb{P}(x,y \mid \mathsf{A}) = \frac{p_{xy}\, \mathbb{P}(x,y)}{\sum_{x',y'} p_{x'y'}\, \mathbb{P}(x',y')}, \tag{C1}$$

and the conditional expected utility is

$$\mathbb{E}\big[u_i \,\big|\, \mathsf{A}\big] = \sum_{x,y} \mathbb{P}(x,y \mid \mathsf{A})\, u_i(x,y). \tag{C2}$$

For the Newcomb payoffs (14), conditional scoring restores dominance entrywise:

$$\begin{aligned} u_1(D,C) - u_1(C,C) &= (M+m) - M = m,\\ u_1(D,D) - u_1(C,D) &= m - 0 = m. \end{aligned} \tag{C3}$$

Thus D strictly dominates C in the conditional-on-success normal form, independently of θ. Per-attempt valuation keeps the survival factors p_{xy} attached to the payoffs; conditional-on-success valuation renormalizes them away.

The entrywise dominance just computed holds with the prior-record variable Y held fixed. The policy-level conditional expected utility $\mathrm{U}_{\mathsf{A}}(x) := \mathrm{E}[u_1 \mid \mathsf{A}, X = x]$ integrates over Y against $\Pr(Y \mid \mathsf{A}, X = x)$, and the posterior depends on x because A couples X and Y through the success weights:

$$\Pr(Y = y \mid \mathsf{A}, X = x) = \frac{p_{xy}\, \Pr(Y = y)}{\sum_{y'} p_{xy'}\, \Pr(Y = y')}. \tag{C4}$$

For the Newcomb payoffs at the unbiased setup prior $\beta = 1/2$, $\mathrm{U}_{\mathsf{A}}(C) = M\Gamma/(\Gamma+1)$ and $\mathrm{U}_{\mathsf{A}}(D) = (M + m + m\Gamma)/(\Gamma+1)$, so

$$\mathrm{U}_{\mathsf{A}}(D) - \mathrm{U}_{\mathsf{A}}(C) = \frac{(M+m) - \Gamma(M-m)}{\Gamma+1}, \tag{C5}$$

and the policy comparison favors one-boxing for $\Gamma > (M+m)/(M-m)$, the risk-dominance boundary of Sec. IV.

For general $\beta \in (0,1)$,

$$\begin{aligned} \mathrm{U}_{\mathsf{A}}(C) &= \frac{\beta\Gamma M}{\beta\Gamma + 1 - \beta},\\ \mathrm{U}_{\mathsf{A}}(D) &= \frac{\beta(M+m) + (1-\beta)\Gamma m}{\beta + (1-\beta)\Gamma}. \end{aligned} \tag{C6}$$

Thus $\mathrm{U}_{\mathsf{A}}(C) > \mathrm{U}_{\mathsf{A}}(D)$ iff

$$\begin{aligned} \beta(1-\beta)(M-m)\Gamma^2 + m\big(2\beta(1-\beta) - 1\big)\Gamma&\\ - \beta(1-\beta)(M+m) > 0.& \end{aligned} \tag{C7}$$

At $\beta = 1/2$ this reduces to $\Gamma > (M+m)/(M-m)$, the risk-dominance boundary. As $\beta \to 1$, the comparison tends to the fixed-record entrywise result M versus $M+m$, so conditional scoring favors two-boxing.

The entrywise statement is therefore a feature of fixed-Y accepted branches; the policy comparison that the agent actually faces is prior-dependent.

Appendix D: Payoff-ordering classification

This appendix states and proves the payoff-ordering proposition of Sec. IV, with entries (24).

Proposition (Structure and payoff-ordering boundaries)**.** *Let $M > m > 0$, let $0 < p_{\text{match}}, p_{\text{mismatch}} < 1$, define $\Gamma := p_{\text{match}}/p_{\text{mismatch}}$, and score utilities per attempt. The postselected agreement filter induces the symmetric normal form with $R = p_{\text{match}}M$, $T = p_{\text{mismatch}}(M+m)$, $P = p_{\text{match}}m$, and $S = 0$. Then:*

(i) *The symmetrized table has strict Prisoner's-Dilemma ordering, $T > R > P > S$, iff $\Gamma < (M+m)/M$.*

(ii) *The symmetrized table has the Stag-Hunt/assurance coordination structure, $R > T$, $R > P$, and $P > S$, iff $\Gamma > (M + m)/M$.*

(iii) *Within this coordination region, the strict ordinal Stag-Hunt chain $R > T > P > S$ holds iff $(M + m)/M < \Gamma < (M + m)/m$.*

(iv) *For $\Gamma > (M + m)/m$, the ordinal ordering is $R > P > T > S$. This is a strong-assurance, or diagonal-dominant, coordination regime: the fixed-point structure is unchanged, but mutual defection exceeds the unilateral temptation payoff per attempt.*

(v) *At $\Gamma = (M + m)/M$ one has $R = T$, so D weakly dominates C and (C, C) is a weak fixed point. At $\Gamma = (M+m)/m$ one has $T = P$, so the ordinal chain degenerates to $R > P = T > S$; the temptation payoff per attempt has fallen to the level of mutual defection.*

(vi) *Within the coordination region, (C, C) is risk-dominant iff $R - T > P - S$, equivalently $\Gamma > (M + m)/(M - m)$.*

In the limiting endpoint cases, the normal form degenerates further: $R = P = S = 0$ when $p_{\text{match}} = 0$, and $T = S = 0$ when $p_{\text{mismatch}} = 0$.

Proof. Since $M > m > 0$ and $p_{\text{match}}, p_{\text{mismatch}} > 0$, we have $R > P > S$ and $T > S$. The comparison between R and T gives $R > T$ iff $p_{\text{match}}M > p_{\text{mismatch}}(M + m)$, iff $\Gamma > (M+m)/M$. This is the boundary between Prisoner's-Dilemma ordering and the Stag-Hunt/assurance coordination structure. The remaining ordinal comparison is $T > P$, which holds iff $p_{\text{mismatch}}(M + m) > p_{\text{match}}m$, iff $\Gamma < (M + m)/m$. Combining these inequalities gives the strict ordinal Stag-Hunt chain $R > T > P > S$; reversing the second gives $R > P > T > S$. Finally, $R - T > P - S$ iff $p_{\text{match}}M - p_{\text{mismatch}}(M + m) > p_{\text{match}}m$, iff $p_{\text{match}}(M - m) > p_{\text{mismatch}}(M + m)$, iff $\Gamma > (M + m)/(M - m)$. □

Appendix E: Potential landscape and basins

This appendix describes the potential, basins, and barrier for the symmetric two-player reading in which Y is a second decision register; in the single-agent reading they are the structure the setup prior β inherits. Writing $x \in [0, 1]$ for the one-boxing weight of the symmetric profile, the selection gradient $U(C \mid x) - U(D \mid x) = \Delta\, x - (P - S)$ integrates to the potential

$$\Phi(x) = \tfrac{1}{2}\,\Delta\, x^2 - (P - S)\, x, \qquad \Delta := R - S - T + P, \tag{E1}$$

which under replicator dynamics is climbed according to $\dot{\Phi} = x(1 - x)\,[\,U(C \mid x) - U(D \mid x)\,]^2 \geq 0$, so Φ is a Lyapunov function for that flow. In the coordination region $\Delta > 0$ the two pure profiles $x = 0$ (two-boxing) and $x = 1$ (one-boxing) are the attractors, separated by the interior repeller $x^* = (P - S)/\Delta = \beta_{\text{ind}}$ of Eq. (27). Reading $V := -\Phi$ as a potential landscape gives a double well whose minima are the two equilibria and whose barrier sits at β_{ind}; with the Newcomb entries (24) the barrier height above the two-boxing well is

$$V(x^*) - V(0) = \frac{P^2}{2\Delta} = \frac{p_{\text{match}}^2\, m^2}{2\,(M + m)\,(p_{\text{match}} - p_{\text{mismatch}})}, \tag{E2}$$

fixed in shape by Γ and m/M. The one-boxing equilibrium is the global maximizer of Φ (equivalently the deeper well of V) iff $\Phi(1) > \Phi(0)$, i.e. $\Delta > 2P$, i.e. $R - T > P - S$: the risk-dominance condition (28), $\Gamma > (M + m)/(M - m)$. The risk-dominance boundary is therefore the value of Γ at which the global minimum of the landscape switches wells.

Appendix F: Side-by-side decomposition

This appendix states and proves the side-by-side proposition of Sec. V.

Proposition (side-by-side decomposition and the PD→SH transition). *Let $M > m > 0$ and score utilities per attempt.*

(i) *For every $\Gamma \geq 1$, read parametrically, each player's decision in $G(\Gamma)$ is the standard Newcomb problem with payoffs u_1 and predictor reliability $q = \Gamma/(\Gamma + 1)$: two-boxing dominates entrywise, $u_1(D, y) - u_1(C, y) = m > 0$, while one-boxing carries the larger expected utility iff $q > (M + m)/2M$, i.e. $\Gamma > (M + m)/(M - m)$.*

(ii) *Read as a game, $G(\Gamma)$ is Prisoner's-Dilemma–ordered for $\Gamma < (M + m)/M$ and Stag-Hunt–ordered for $\Gamma > (M + m)/M$, with (C, C) risk-dominant iff $\Gamma > (M + m)/(M - m)$ (Prop. D).*

(iii) *The thresholds of (i) and (ii) coincide: each side's Newcomb verdict turns to one-boxing at the same $\Gamma = (M + m)/(M - m)$ at which (C, C) becomes risk-dominant in the composite.*

Hence $G(\Gamma)$ is two Newcomb problems side by side, in Lewis's sense, throughout the predictor regime $\Gamma \geq 1$; the consistency factor changes only the equilibrium structure of the game they compose into. The crossover at $\Gamma = (M + m)/M$ is a transition of a genuine two-player game built from two Newcomb components, carrying it from Prisoner's-Dilemma to Stag-Hunt ordering.

Proof. (i) Holding the prediction x_j fixed, the raw comparison is $u_1(D, y) - u_1(C, y) = m > 0$. With reliability q, EU(one-box) $= Mq$ and EU(two-box) $= M + m - Mq$ (Sec. III D), so one-boxing wins iff $q > (M + m)/2M$; substituting $q = \Gamma/(\Gamma + 1)$ gives $\Gamma > (M + m)/(M - m)$. (ii) is Prop. D. (iii) equates the two thresholds. □

Appendix G: Forecasting-predictor optimum

With W as defined in Eq. (29), under the best response (31), the agent's reduced payoff is $W(\alpha,1) = \alpha R + (1-\alpha)T$ for $\alpha > \frac{1}{2}$ and $W(\alpha,0) = \alpha S + (1-\alpha)P$ for $\alpha < \frac{1}{2}$, each affine in α. The lower branch is decreasing (since $P > S$), with maximum P at $\alpha = 0$. The upper branch has slope $R - T$. In the Stag-Hunt region $R > T$ it is increasing, so its maximum is R at $\alpha = 1$; because $R > P$ always, the global optimum is the pure corner $\alpha^*_{\mathrm{MAP}} = 1$, Eq. (32).

In the Prisoner's-Dilemma region $R < T$ the upper branch $W(\alpha,1)$ decreases in α, so the agent prefers α just above $\frac{1}{2}$. The payoff approaches $(R+T)/2$ as $\alpha \to \frac{1}{2}^+$ but no preparation attains it, since at $\alpha = \frac{1}{2}$ the maximum-a-posteriori tie-break flips β to the opposite corner; optimizing against this threshold discontinuity yields a supremum rather than a maximum. The coordination region is clean for the complementary reason: there one-boxing and being forecast as a one-boxer align, so the optimum is the pure corner $\alpha = 1$.

Appendix H: Self-calibrated predictor

A calibrated predictor matched to the realized accepted-run frequency sets $\beta = \Pr(X{=}C \mid \mathsf{A})$. With $\delta := p_{\mathrm{match}} - p_{\mathrm{mismatch}}$, $w_C(\beta) = p_{\mathrm{mismatch}} + \delta\beta$, $w_D(\beta) = p_{\mathrm{match}} - \delta\beta$, the condition is linear in α and inverts as

$$\alpha(\beta) = \frac{\beta\, w_D}{Z}, \quad 1-\alpha(\beta) = \frac{(1-\beta)\, w_C}{Z}, \qquad Z = p_{\mathrm{mismatch}} + 2\delta\beta(1-\beta). \tag{H1}$$

Since $\mathrm{d}\alpha/\mathrm{d}\beta = p_{\mathrm{mismatch}}\left[p_{\mathrm{match}} - 2\delta\beta(1-\beta)\right]/Z^2 > 0$, the map is an increasing bijection of $[0,1]$ and the fixed point is unique, with $\alpha = 1 \Leftrightarrow \beta = 1$. Along this curve the per-attempt payoff (29) is

$$W(\beta) = \frac{\beta^2 w_D R + \beta(1-\beta) w_C T + (1-\beta)^2 w_C P}{Z}, \tag{H2}$$

and a direct computation gives

$$R - W(\beta) = \frac{(1-\beta)\, Q(\beta)}{Z}, \tag{H3}$$

with Q quadratic, leading coefficient $\delta\left[\, m(p_{\mathrm{match}} - p_{\mathrm{mismatch}}) - M(p_{\mathrm{match}} + p_{\mathrm{mismatch}})\,\right] \leq 0$ for $\Gamma \geq 1$, so Q is concave on $[0,1]$. Its endpoints, $Q(0) = p_{\mathrm{match}} p_{\mathrm{mismatch}}(M - m) > 0$ and $Q(1) = p_{\mathrm{match}}(M p_{\mathrm{match}} - m\, p_{\mathrm{mismatch}}) > 0$, are positive; concavity then gives $Q > 0$ throughout, so $R - W(\beta) \geq 0$ with equality iff $\beta = 1$. Hence the agent's optimum is $\alpha^* = 1$ with payoff R for every $\Gamma \geq 1$, matching the verdict under the $\beta = \alpha$ predictor. The two notions of calibration differ only in the off-optimum response.

Appendix I: Adversarial adaptive preparation

This appendix gives the lower-envelope calculation quoted in Sec. VI. Here W is the payoff in Eq. (29) and $\Delta = R - S - T + P$ as in Eq. (30). The adversarial objective lets the predictor choose the setup prior $\beta \in [0,1]$ after observing the Born weight α, minimizing the agent's per-attempt payoff:

$$V_{\mathrm{adv}}(\alpha) := \inf_{\beta \in [0,1]} W(\alpha, \beta).$$

Since W is affine in β, with

$$\frac{\partial W}{\partial \beta} = T - P + \alpha\Delta,$$

the minimizing prior lies at one of the two corners of the interval. In the predictor regime $\Gamma \geq 1$, the Newcomb entries give $\Delta \geq 0$. If $T \geq P$, then $\partial W/\partial\beta \geq 0$ for every α, so the minimizing prior is $\beta = 0$. Hence

$$V_{\mathrm{adv}}(\alpha) = W(\alpha, 0) = \alpha S + (1-\alpha)P.$$

For the Newcomb matrix (24), $P > S$, so this lower envelope is maximized at $\alpha = 0$. The robust adaptive preparation is therefore pure two-boxing whenever $T \geq P$. At $\Gamma = 1$, the filter has no consistency bias, $\Delta = 0$, and $T - P = p_{\mathrm{match}} M > 0$, so the same conclusion holds.

If $T < P$, the lower envelope has an interior kink, determined by

$$W(\alpha, 0) = W(\alpha, 1).$$

Using Eq. (30), this gives

$$\alpha^* = \frac{P - T}{\Delta}. \tag{I1}$$

The dual prior that makes the agent indifferent satisfies

$$\frac{\partial W}{\partial \alpha} = 0,$$

hence

$$\beta^* = \frac{P - S}{\Delta}. \tag{I2}$$

For the Newcomb entries (24), the condition $T < P$ is

$$p_{\mathrm{mismatch}}(M + m) < p_{\mathrm{match}} m, \qquad \Gamma > \frac{M + m}{m}. \tag{I3}$$

Substituting the Newcomb entries into Eqs. (I1)–(I2) gives

$$\alpha^* = \frac{P - T}{\Delta} = \frac{\Gamma m - (M + m)}{(M + m)(\Gamma - 1)}, \tag{I4}$$

$$\beta^* = \frac{P - S}{\Delta} = \frac{\Gamma m}{(M + m)(\Gamma - 1)}. \tag{I5}$$

The setup prior β^* coincides with the fixed-β basin boundary β_{ind} in Eq. (27); the quantity α^* is the Born weight of one-boxing in the lower-envelope adaptive preparation. The two quantities are generally distinct:

$$\beta^* - \alpha^* = \frac{1}{\Gamma - 1}.$$

This gap closes as Γ grows, and at $\Gamma \to \infty$ both reach

$$\frac{m}{M+m}.$$

Thus the adversarial robust preparation remains overwhelmingly two-boxing in the standard Newcomb regime $M \gg m$, even in the ideal consistency limit.

Appendix J: Complementarity and the σ_x signature

This appendix proves the distinguishability–visibility bound of Sec. VII and its corollary, and gives the symmetric-preparation state algebra behind $\langle \sigma_x \rangle_X$.

Proof of the bound (37). The setup is a which-path experiment: the act X is the path, the record system Y is the detector monitoring it, and the reliability $q = \Pr(y = x)$ is the probability of guessing the path from the detector readout. The inequality is then the wave–particle duality relation for Y monitoring X. The agreement-based quantity $\mathcal{D} = 2q-1$ is the record-conditioned distinguishability, bounded by the optimal which-way distinguishability $\mathcal{D}_\star$ extractable from Y, $\mathcal{D} \le \mathcal{D}_\star$ [29]. Englert's relation bounds the distinguishability against the phase-optimized transverse visibility, $\mathcal{D}_\star^2 + \mathcal{V}^2 \le 1$ [28], so $\mathcal{D}^2 + \mathcal{V}^2 \le 1$. When the record measurement is optimal and the relevant quanton–record state is pure, the standard pure-detector case gives equality. This is the case realized by the coherent parity filter (Sec. VII). □

Proof of the no-early-fact corollary. $\Pr(y = x) = 1$ with a designated computational-basis readout of Y means the joint distribution of the X-readout and the Y-readout is supported on the diagonal. The joint state may still carry coherences inside that diagonal subspace, but it has the form

$$\rho_{XY} = \sum_{y,y'} r_{yy'} \, |y\rangle_X |y\rangle_Y \langle y'|_X \langle y'|_Y$$

for some matrix $r_{yy'}$. Tracing over Y removes the $y \neq y'$ terms, so

$$\rho_X = \sum_y r_{yy} \, |y\rangle\langle y|$$

is diagonal and $\mathcal{V} = 0$. A perfect classical readout of X is created by a $\{C, D\}$-basis measurement, which leaves the diagonal Born weights untouched and removes only the off-diagonal coherence, so $p_C = \alpha$. The endpoint $\Gamma \to \infty$ of (38) realizes this, with $\mathcal{D} \to 1$, $\mathcal{V} \to 0$. □

The σ_x state algebra. The two CNOTs from X and Y into the ancilla map $|+\rangle_X \otimes |+\rangle_Y \otimes |0\rangle_a$ to

$$\tfrac{1}{2}\big(|00\rangle\,|0\rangle + |01\rangle\,|1\rangle + |10\rangle\,|1\rangle + |11\rangle\,|0\rangle\big)_{XYa}, \qquad \text{(J1)}$$

with the ancilla carrying the parity $x \oplus y$. The rotation $R_y(\theta)$ on the ancilla followed by postselection on $|0\rangle_a$ yields the normalized joint state

$$|\psi(\theta)\rangle_{XY} = \tfrac{1}{\sqrt{2}}\big(c\,|00\rangle - s\,|01\rangle - s\,|10\rangle + c\,|11\rangle\big), \qquad \text{(J2)}$$

with $c = \cos(\theta/2)$ and $s = \sin(\theta/2)$. At the symmetric preparation each parity sector has prior weight $1/2$, so the acceptance probability is $(c^2+s^2)/2 = 1/2$, independent of θ. Within the accepted ensemble the match and mismatch probabilities are c^2 and s^2, recovering $\Gamma = \cot^2(\theta/2)$.

The reduced state on X is

$$\rho_X(\theta) = \tfrac{1}{2}\big(I - \sin\theta \cdot \sigma_x\big) = \tfrac{1}{2}\left(I - \frac{2\sqrt{\Gamma}}{\Gamma+1}\sigma_x\right), \qquad \text{(J3)}$$

so $\langle \sigma_x \rangle_X = -2\sqrt{\Gamma}/(\Gamma+1)$. The sign is negative because agreement-favoring postselection retains the branch whose Kraus amplitude on the mismatch sector carries the relative minus sign of Eq. (7), imprinting an anti-aligned σ_x coherence. The classical rejection-sampling model gives $\rho_X^{\mathrm{cl}} = I/2$ and $\langle \sigma_x \rangle_X^{\mathrm{cl}} = 0$ at every Γ.

The local signature depends on the implementation phase: for the symmetric phase family $K = c\,\Pi_= + e^{i\varphi} s\,\Pi_{\neq}$ the local σ_x signal is proportional to $\cos\varphi$, vanishing at $\varphi = \pi/2$ rather than rotating into a σ_y signal, so the real-phase $R_y(\theta)$ circuit maximizes it. The discriminator is accessible in a two-qubit linear-optical setup of the kind demonstrated for P-CTC postselection [14]: prepare both registers in $|+\rangle$, apply the parity-filter circuit of Fig. 1, herald on the ancilla outcome $|0\rangle$, and measure σ_x on the act register. Resolving $|\langle \sigma_x \rangle_X|$ to statistical precision ϵ takes of order ϵ^{-2} heralded events, the $1/2$ heralding rate doubling the raw runs.

Appendix K: Maximal-ignorance limit

The maximally noisy case of Sec. VIII is θ uniform on $[0, \pi]$. The angle is the natural ignorance variable: it parameterizes the rotation gate directly, and the uniform prior is symmetric about $\theta = \pi/2$, so the averaged survival weights coincide; a flat prior in Γ or in p_{mismatch} would privilege one sector by construction. Run by run, the realized filter is Stag-Hunt-ordered for $\theta < \theta^* \equiv 2\arctan\sqrt{M/(M+m)}$, so $\Pr(\mathrm{SH}) = \theta^*/\pi$ and $\Pr(\mathrm{PD}) - \Pr(\mathrm{SH}) = 1 - 2\theta^*/\pi = m/(\pi M) + O(m^2/M^2)$: the hidden split is close to even, with a small Prisoner's-Dilemma excess set by the transparent-box payoff.

The ordering of the averaged channel is not the run-by-run majority, because the payoff ordering is not linear in the sector weights. For θ uniform, $\mathbb{E}[\cos^2(\theta/2)] = \mathbb{E}[\sin^2(\theta/2)] = \frac{1}{2}$, so $\Gamma_{\mathrm{eff}} = 1$: the averaged game is Prisoner's-Dilemma-ordered, even though half the individual runs land on the Stag-Hunt side. At $\Gamma_{\mathrm{eff}} = 1$ the filter supplies no consistency bias, agreement and disagreement surviving equally, and the model carries no match-mismatch signal. The averaged game is Stag-Hunt-ordered only when $\mathbb{E}[\cos^2(\theta/2)] > (M+m)/(2M+m)$.

The agent, lacking the realized θ, sees only E_{eff}, and its payoff is governed by Γ_{eff}.

Appendix L: Process-matrix form and causal separability

This appendix casts the $\beta = 1/2$ map as a process matrix W and asks whether the construction draws on its causal order. A causally nonseparable W would mean it does; the calculation returns a classical common cause, diagonal in the computational basis and causally separable, with no signaling between the registers. The only quantum content is then the local σ_x coherence of Sec. VII A, with none in the causal structure of W.

A postselected map is a process matrix when the acceptance probability is fixed across the local operations, so that the renormalization $\rho \mapsto K\rho K^\dagger / \operatorname{Tr}[K\rho K^\dagger]$ stays linear [45]; the per-attempt numerator is linear regardless. That probability is $\Pr(\mathsf{A} \mid \alpha, \beta)$ of Theorem 1(i), with $\partial \Pr(\mathsf{A})/\partial\alpha = (p_{\text{match}} - p_{\text{mismatch}})(2\beta - 1)$, vanishing for all α iff $\beta = 1/2$. The unbiased prior is the process-matrix locus; for $\beta \neq 1/2$ the map is a general nonlinear P-CTC, with the Simpson reversal of Sec. IX B as its signature.

Read the two registers as parties A and B, each a closed laboratory with input and output, in the side-by-side decomposition of Sec. V. At $\beta = 1/2$ the accepted ensemble correlates the two through agreement at rate $q = \Gamma/(\Gamma + 1)$ (Sec. II D). This correlation is generated by the process matrix

$$W = \tfrac{1}{4}\big(I + \mathcal{D}\,\sigma_z^{A_I} \otimes \sigma_z^{B_I}\big) \otimes I^{A_O B_O}, \qquad \mathcal{D} = \frac{\Gamma - 1}{\Gamma + 1}, \tag{L1}$$

with $\mathcal{D}$ the distinguishability of Eq. (12). Equation (L1) is a valid process matrix: it is positive, has $\operatorname{Tr} W = d_{A_O} d_{B_O} = 4$, and lies in the valid subspace,

$$W = {}_{A_O}W + {}_{B_O}W - {}_{A_O B_O}W, \qquad {}_X W := \tfrac{1}{d_X} I_X \otimes \operatorname{Tr}_X W. \tag{L2}$$

It places the record in the shared past of the two parties, the common-cause reading of Fig. 2(a); its time reverse is the collider of Fig. 2(b). The common-cause drawing applies because the $\beta = 1/2$ map carries no signaling between the parties, and the retrocausal reading of Fig. 2(c) is the same correlation with the act edge reversed. Identifying the process matrix both as the linear P-CTC and as the common cause is then consistent: linearity is a property of the map, while (a) and (c) are two of Reichenbach's explanations of the one no-signaling correlation it carries. Born's rule $p = \operatorname{Tr}[W(M_A \otimes M_B)]$ with computational-basis readouts returns agreement q.

The process is causally separable. It carries no signaling between A and B, so it admits both orders $A \prec B$ and $B \prec A$ and is a fixed-order resource in either. The Kraus operator (7) fixes the classical character: its Pauli support is $I \otimes I$ and $\sigma_z \otimes \sigma_z$, the diagonal correlation of a common cause, with no $\sigma_z \otimes \sigma_x$ cross-register term. A bipartite process that violates a causal inequality requires such a term [46], and classical operations reach causal nonseparability only with three or more parties, the regime of the Aráujo–Feix–Baumeler–Wolf process [47]. The two-party object here does not reach it: the deterministic agreement loop $a_I = b_O,\ b_I = a_O$ fails Eq. (L2), so no consistent cyclic bipartite process underlies the filter, and the postselection supplies the consistency a third register would otherwise carry. The loop is the non-uniquely-solvable identity two-cycle, to which Ferradini, Gitton and Vilasini [50] assign the uniform agreement distribution $\delta_{ab}/2$ by classical postselected teleportation, the distribution the filter itself returns as $\Gamma \to \infty$. The local σ_x coherence of Sec. VII A certifies coherent enforcement and does not enter the causal structure of W.

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